Defect Trigger Identification in a Construction Digital Twin Using Quality Linked Data and Iterative Deviation Computation

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Abstract: Current quality management in construction faces several challenges, including insufficient precision in control, unclear allocation of quality responsibilities, and a lack of long-term management mechanisms. As the industry shifts from a problem-solving after occurrence approach to a proactive paradigm, identification of potential quality defects and risks has become a primary prerequisite. This study introduces the concept of quality defect triggers, detectable, modelable, and controllable precursors to quality issues that operationalize risk factors and can be visualized within a construction digital twin (DT). Building on an exploration of their evolutionary mechanisms, the study develops an integrated framework that combines ontology and linked data technologies to identify quality defect triggers through multisource heterogeneous data integration, comprising three core components: (1) the evolutionary mechanisms of defect trigger as the theoretical foundation and underlying principles for computation; (2) an ontology-based four-layer semantic model encompassing quality standards, quality data, defect triggers, and construction processes, implemented using the GraphDB graph database to achieve semantic linkage; and (3) a quality defect trigger identification and iterative deviation computation engine in construction DT. By unifying data modeling, semantic mapping, and automated reasoning, the framework provides a systematic and practical approach to early quality defect identification. Its effectiveness was validated through a real-world prefabricated composite-slab installation case, comparative experiments, and expert evaluation, demonstrating its applicability and potential to support proactive, data-informed quality management in construction practice.
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Introduction

Over the last decade, the architecture, engineering, construction, and operations (AECO) industry has made substantial investments in digital transformation. Building information modeling (BIM) has become the mainstream modeling tool, Internet of Things (IoT) sensors continuously capture on-site process data, and digital twin (DT) technologies are increasingly used for visualization of geometry, progress, and operations (Afzal et al. 2023; Liu et al. 2024b).

Governments and industry associations in various countries have successively issued digital delivery and information management guidelines to promote data collection and sharing throughout the entire project life cycle. For example, the Ministry of Housing and Urban-Rural Development of the People's Republic of China (2022) introduced the *14th Five-Year Plan for the Development of the Construction Industry*, and a German-based consulting firm (Roland Berger 2016) published the report *Digitization in the Construction Industry: Building Europe's Road to "Construction 4.0,"* which outlines the digital transformation of the construction sector.

However, multiple international surveys have indicated that the direct cost of rework still accounts for 11%–15% of total project investment, while claims and schedule delays caused by quality defects are even on the rise (Love et al. 2009; Olanrewaju 2025). This phenomenon highlights that despite the growing availability of digital tools and the accumulation of quality data, construction quality issues have not declined.

Although current quality management practices incorporate risk control measures (risk identification, inspections, sampling, and approvals), significant gaps persist in linking abstract risks to concrete events, tracking the rapid evolution of risks, and ensuring the implementation of proactive measures (Guo et al. 2025; Jeon et al. 2023; Mohamad and Tran 2021). Quality management benefits from the mature domain of safety in construction, which distinguishes among risk, hazard, and accident and develops proactive identification methods (Chen et al. 2025b; Li et al. 2018; Pereira et al. 2018; Yang et al. 2012). However, unlike safety, quality is an intrinsic project attribute that extends beyond construction completion or handover (Bowen et al. 2002; Mallawaarachchi and Senaratne 2015).

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Due to its long duration and complexity, the mechanism of quality defect formation and evolution does not fully align with the hazard–accident discontinuous mechanism in the safety field. The formation of defects may involve a long time span and be influenced by the cumulative effects of quality deviations, while also being triggered by various events. This includes both the risk–incident disruptive effect (He and Atangana Njock 2023; Okonta et al. 2024) and the risk–(deviation/defect trigger)–defect gradual effect (Faqih and Zayed 2021; Ma et al. 2021). The former has a severe impact on quality incidents and safety, and the latter affects quality attributes over the long term, extending into the operational phase and throughout the building's life cycle. However, this coupled mechanism is underexplored in existing research, and current quality hazard inspections in practice, being experience-driven and checklist-based, struggle to capture finer-grained hazard elements, including defect triggers and deviations.

Although current individual innovations and technologies, such as defect detection (Guo et al. 2021; Liang et al. 2023), quality data integration (Kovacs and Micsik 2021; Yan et al. 2025b), and design specification checks (Ma et al. 2024; Sydora and Stroulia 2020), are well-developed and capable of addressing existing quality issues, they often are not related to preventive measures and interpretability. These approaches tend to emphasize the identification of issues at later stages rather than addressing the underlying latent factors that could trigger defects.

Technically, the challenge lies in developing methods that can capture proactive measures and provide the necessary interpretability for early-stage quality hazard identification in construction projects. This requires addressing the complexities of integrating diverse data sources, such as model files [(Industry Foundation Class (IFC)/BIM metadata], real-time sensor readings (e.g., concrete strength, temperature–humidity, vibration records), and inspection reports (PDF scans of concrete-cube tests and rebar reinspection results) to event and decision logs (modification orders, and image records) (Choi and Lee 2022; Iqbal et al. 2024; Tuhaise et al. 2023). Current DT development provides a solid foundation for integrating and presenting both structured and unstructured quality data with model views (Cai et al. 2021; Liu et al. 2024b; Ramonell et al. 2023; Yan et al. 2025b), and can incorporate checking algorithms to simulate the accumulation of deviations and improve the early-stage identification of defects.

Therefore, this study focuses on investigating the underlying mechanisms of quality defects and adapting application technologies to develop new methods, aiming to achieve the following objectives:

1. Define the concept of quality defect triggers, explore the risk–(deviation/defect trigger)–defect–incident evolutionary mechanism with an emphasis on the precursory nature of defect triggers and deviations, and clarify their identification characteristics at a finer granularity.
2. Develop a hierarchical framework that integrates construction processes, quality standards, quality data, and quality defect triggers, and apply ontology-based approaches to model their relationships, ensuring contextual adaptation.
3. Integrate multisource heterogeneous quality data and construct quality linked data (QLD) for identifying defect triggers within the context of construction DT, facilitating more precise and contextually relevant identification.

The remainder of this paper is organized as follows. Section "Related Work" reviews the latest research progress on quality defect trigger identification and highlights the research gaps that this study aims to address. Section "Research Methodology" presents the methodology proposed in this paper. Section "Framework of Quality Linked Data for Defect Trigger Identification" elaborates on the overall framework, key functional modules, and technical details of the defect trigger identification framework based on QLD. Section "Implementation and Validation" presents a case study on the prefabricated composite-slab construction process to evaluate the functional feasibility, effectiveness, and practical value of the proposed framework in engineering practice, based on comparative experiments and expert interviews. Section "Discussion" discusses the theoretical contributions and practical implications of this study. Finally, section "Conclusion" summarizes the paper and outlines directions for future research.

Related Work

Quality Issue Evolution Mechanism and Defect Trigger Identification

In the construction field, quality management can draw on the more mature domain of safety management, which offers analytical perspectives for distinguishing among risk, hazard, and accident. According to reviews on safety management in construction (Alkaissy et al. 2020; Sousa et al. 2014; Zhou et al. 2015), early safety research began with accident analysis studies that examined numerous accident cases and developed causation models (e.g., Abdelhamid and Everett 2000; Suraji et al. 2001) and statistical analyses (e.g., Huang and Hinze 2003; Shin et al. 2018). Subsequently, the focus shifted to accident prevention studies such as design for safety (e.g., Behm 2005; Yang et al. 2012), and later progressed toward risk evaluation research (Li et al. 2017; Lu et al. 2021).

Previous construction safety studies have recognized two stages in elaborating the mechanisms underlying the evolution of safety issues. The first stage, i.e., the context-general stage, conceptualizes the relationships among risk, hazard, and accident at a macro level. Early safety research, based on accident investigations, sought to develop generalizable risk management theories (Patterson and Neailey 2002). Here, risk precedes hazard, defined abstractly as the uncertainty affecting objectives, a latent indicator of potential hazards (Albert et al. 2020), consistent with general risk management standards such as ISO 31000:2009 (ISO 2009) (Sui Pheng and Wee 2001). The second, context-specific stage, defines these relationships at a micro level, emphasizing how specific hazards and risks emerge within project environments. In this stage, hazard precedes risk, as identifiable domain hazards (Carter and Smith 2006; Perlman et al. 2014) are used to assess risk through probability and consequence severity.

Unlike progress or safety, quality is an intrinsic attribute of a project that does not end with the completion of construction or the handover of the project. Because of its long time span and the complexity of influencing factors, the current mechanism of quality defect identification is more analogous to the first stage in safety research. Fig. 1 illustrates the underlying mechanism of quality issue evolution, developed with reference to the mature frameworks established in safety management (Carter and Smith 2006).

The evolution mechanism among quality risk, hazard, defect, and incident is not linear, as illustrated in Fig. 1. Nevertheless, this study uses risk–(deviation/defect trigger)–defect–incident to represent it conceptually. According to the mechanism established in safety research, hazards in the quality domain can be categorized into consequence-oriented and process-oriented types, i.e., deviation and defect triggers. The deviation that occurs during quality transmission partially corresponds to a near miss in safety studies (Carter and Smith 2006). When it exceeds standard requirements, it is classified as a quality failure (Barber et al. 2000); when it does not exceed but may cause potential impact, it warrants an early warning (Li et al. 2025; Yao et al. 2021). In contrast, a quality defect trigger refers more to an event-type occurrence or the underlying condition that gives rise to it, encompassing nonconforming states,

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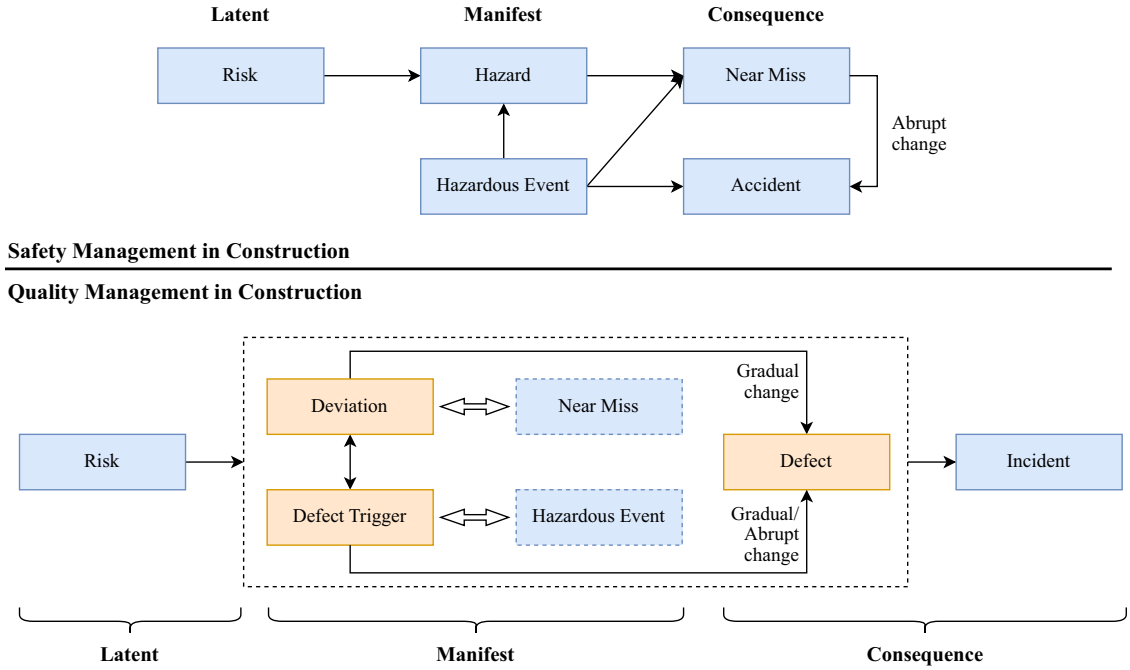


Fig. 1. Quality issue evolution mechanisms in construction compared with safety research.

noncompliant processes, unsafe behaviors, flawed construction activities, managerial deficiencies, or force majeure, corresponding to a hazardous event in the safety domain (Albert et al. 2014; Chang et al. 2020).

Existing techniques and approaches such as knowledge modeling, data integration, and rule-based checking have established a foundation for construction quality issue preventive identification in three dimensions: (1) a knowledge and data foundation for identifying hazards at the granularity of deviations and defect triggers (Chen and Luo 2014; Liu et al. 2024b; Sheng et al. 2020); (2) a methodological foundation for aggregating and validating quality-related data with higher precision thresholds (Kovacs and Micsik 2021; Ma et al. 2024; Yan et al. 2025b); and (3) an environmental foundation for representing the virtual-physical scenario with model views in which deviations and defect triggers emerge (Cai et al. 2021; Liu et al. 2021; Zheng et al. 2020).

Knowledge Modeling and Data Integration

In the AECO industry, knowledge modeling and data integration have evolved from early database and expert systems (McGartland and Hendrickson 1985) to more sophisticated and semantically enriched approaches. The adoption of BIM and open standards such as IFC marked a major step toward digital and interoperable project environments, which formalized domain concepts and systematically constructed building data (Antunes et al. 2024; Bellon et al. 2025; Matarneh et al. 2022). In recent years, technologies such as the semantic web, ontologies, linked data, and knowledge graphs, and digital twin have enabled the more effective integration and reuse of heterogeneous data from multiple sources (He et al. 2025; Liu et al. 2024a, b; Yan et al. 2025a; Zhong et al. 2020).

Ontology and knowledge graph technologies have been widely applied to information integration and data sharing (Ramonell et al. 2023), safety management (Fang et al. 2020), quality management (Xiahou et al. 2025), cost control and schedule management (Pfitzner et al. 2025), and intelligent construction equipment (Chen et al. 2023). In construction quality management, ontologies and knowledge graphs play two main roles: they formalize quality standards and managerial experience into reusable knowledge bases. Although these bases are built via semantic modeling, recent studies showed that large language models (LLMs) can extract structured knowledge from engineering texts.

Using prompt engineering and retrieval-augmented generation (RAG), LLMs identify entities, relations, and clause semantics, then align them with knowledge graphs to improve verifiability. This approach has been validated in contract review, safety accident analysis, quality inspection, and engineering Q&A (Isah and Kim 2025; Liu et al. 2025a, b; Zheng et al. 2025). When knowledge bases are integrated with BIM and computer vision technologies, these tools support design model compliance checks (Zhang and El-Gohary 2017), quality management Q&A system development (Chen et al. 2025a), identification of construction quality issues (Lee et al. 2016; Liang et al. 2023), and traceability of quality issues (Jing and Li 2024; Kim et al. 1999).

Linked data is a technological specification used for publishing and connecting structured data on the Web, aiming to enable data to be automatically read, understood, and inferred by machines through standardized methods (Bizer et al. 2023; Mountantonakis and Tzitzikas 2020). The common implementations are Web Ontology Language (OWL) and Resource Description Framework (RDF) (Bizer et al. 2023).

In AECO, it enables interoperability among BIM/GIS models, IoT sensor streams, and documentary artifacts, supporting cross-domain data integration (Pauwels et al. 2017b; Shen et al. 2024). For construction quality management, it is used to integrate heterogeneous quality data, check construction/quality rules, enrich BIM semantics (Bassier et al. 2024; Werbrouck et al. 2020), and link facilities management (FM) systems with BIM components (Kim et al. 2018). Ontologies supply the shared vocabulary, and knowledge graphs instantiate and connect entities, forming the semantic backbone; recent work has moved from ontology construction to BIM integration, intelligent reasoning/analytics, and decision support for construction robotics (Farghaly et al. 2023; Pauwels et al. 2017a).

Graph databases are purpose-built, graph theory–based storage systems for highly connected data. They enable fast traversal,

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relationship queries, and stronger semantic analysis (Angles and Gutierrez 2008; Pokorný 2015; Sellami et al. 2025). Typical graph database tools include Neo4j, GraphDB, Fuseki, and Virtuoso (Dervišević 2021; Li et al. 2022). In AECO, graph databases manage multisource RDF data to support process design, collaboration, scheduling, safety, and operations (Yang et al. 2023; Zhu et al. 2025). In construction quality management, graph databases facilitate model compliance checking and health monitoring (Ramonell et al. 2023; Schuk et al. 2022).

Rule-Based Checking in Construction Digital Twin

Current information modeling techniques in the AECO industry, such as BIM, are predominantly aligned with the object-oriented paradigm, focusing on the structured representation of entities, relationships, and rules through data schemas, typically implemented in shared databases or standardized data formats (Choi et al. 2016; Manzoor et al. 2021). It has enabled the transformation from human-understandable representations of constructed assets to machine-interpretable information, while offering a well-established framework and infrastructure for the storage and retrieval of as-designed data. As a result, a range of methods and tools for verifying as-designed data (e.g., collision checking and code compliance checking) and comparing as-built data with as-designed data (e.g., construction progress updating, safety warning, and quality tracing) have been developed and are increasingly applied in practice (Ma et al. 2023; Sydora and Stroulia 2020; Wang et al. 2016; Zhang et al. 2023, 2020).

Rule-based checking is the underlying method of the aforementioned applications that involves formalizing codes, standards, or heuristic guidelines into machine-executable rule sets, combined with software engines for automated compliance verification. The core process involves developing a rule base that converts quality acceptance standards and industry expert knowledge into executable rules. For instance, Guo et al. (2021) defined a rule database based on facade evaluation standards and used deep learning technology to detect facade quality defects; Völker and Shokouhi (2015) transformed the physical characteristics of concrete honeycomb defects into quantitative thresholds and used multisensor data to assess concrete quality; Kang et al. (2020) encoded practical knowledge of mechanical, electrical, and plumbing (MEP) system installation into parameters for a Scan-to-BIM workflow, employing point cloud scanning technology to identify installation defects.

Rule-based checking in a modeled environment, such as BIM, primarily involves as-designed (virtual) and as-built (real-world) data. The development of construction DT has unlocked the potential to incorporate full life-cycle timelines and multiple dimensions of data, integrated with a multilevel knowledge graph with accumulated instantiations (Listl et al. 2024; Ramonell et al. 2023). Through dynamic mapping and intervention mechanisms via simulation and learning techniques, it has extended the scope and capabilities of current building information modeling. Therefore, rule-based checking can be applied in a construction DT for more precise quality control, integrating data or data structures from the Work Breakdown Structure (WBS) and Work Packages (WP), as well as various ontologies and knowledge graphs (Liu and Lu 2020; Stas and Abrishami 2024).

Research Gaps

Due to established conventions and practices in quality management and defect prevention, deviations can be identified through quality inspections and data comparisons. However, the methods for identifying defect triggers remain less developed. Many studies have focused on the causation of quality defects, which are typically categorized into direct causes and root causes, covering areas such as organizational influences, defect supervision, and behavioral preconditions (e.g., Josephson and Hammarlund 1999; Lambers et al. 2024; Talebi et al. 2021). However, most of these studies are qualitative and statistical analyses, including long-term project research and observation, rather than timely identification methods. For early identification methods, existing approaches often focus on real-time defect detection (e.g., Guo et al. 2021) and on-site inspections with quality hazard checklists (e.g., Choi and Lee 2022), neglecting the underlying evolution mechanism and lacking accurate, preventive interventions.

From a technical perspective, several studies have incorporated emerging innovations into quality hazard management (e.g., Jahanshahi et al. 2013; Zhong et al. 2025), extending the safety-domain logic of hazard–accident transitions to quality management contexts. These studies provide valuable methodological references; however, quality hazard identification is characterized by a gradual and evolutionary process of quality formation, which necessitates a more comprehensive consideration and reflection of elements such as hazardous events (corresponding to defect trigger) and near misses (corresponding to deviation) in the quality management domain.

Therefore, this study, based on the mechanisms of quality formation and evolution, i.e., risk–(deviation/defect trigger)–defect–incident, reconstructs and recombines technical approaches (e.g., quality data integration and checking) and management methods (e.g., process patterns assignment in DT), aiming to improve the granularity of identification, time span, data utilization, and interpretable presentation of potential quality defects.

Research Methodology

The methodology guides the design and validation of a framework for identifying quality defect triggers. Fig. 2 summarizes three stages. Stage 1 conducts a literature review to define the research problem and objectives. Stage 2 develops a DT-enabled conceptual framework based on the physical–data–virtual architecture, comprising (1) a defect trigger evolution mechanism; (2) an identification ontology for consistent semantics; (3) a QLD module that integrates multisource heterogeneous data from the physical layer; and (4) a deviation calculation and defect trigger identification module. Stage 3 implements a prototype and validates it in a prefabricated composite-slab installation case, and semistructured expert interviews then assess the framework's definitions and functions, ensuring practical relevance.

Framework of Quality Linked Data for Defect Trigger Identification

Based on the objectives and identified gaps of this research, a quality defect trigger identification approach based on QLD is proposed (Fig. 3). This approach first clarifies the evolution mechanisms of quality defect triggers, and then constructs the defect trigger identification ontology. The proposed approach enables the conversion of multisource heterogeneous data into structured RDF data, facilitating the mapping of as-built data and criteria data with the ontology. These RDF data support the calculation of quality deviations and the identification of quality defect triggers through rule-verification and the deviation cumulative effect. The approach comprises the following steps: (1) investigating the evolution mechanisms of

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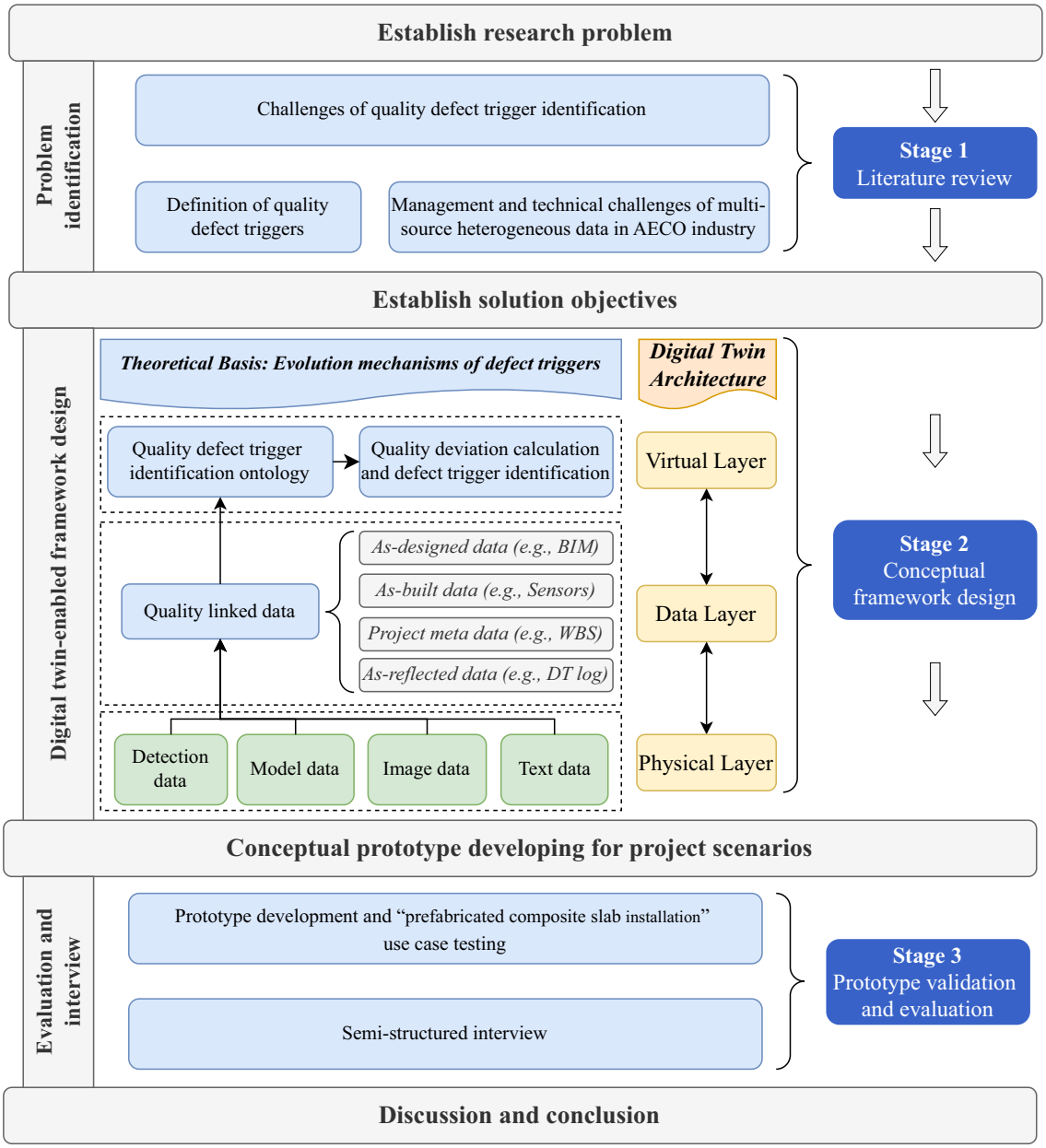


Fig. 2. Research methodology.

quality defect triggers; (2) constructing the quality defect trigger identification ontology; (3) developing QLD for quality deviation calculation; and (4) applying rule-verification and deviation cumulative model for defect trigger identification.

Evolution Mechanisms of Quality Defect Triggers

The evolution mechanisms of defect triggers primarily refer to the process through which defect triggers form and evolve into quality defects and quality accidents. Quality defects and quality accidents are the main manifestations of construction quality nonconformance, with accidents representing a more severe form. Preventing quality defects is essential for achieving high-quality construction outcomes and depends on identifying and eliminating defect triggers.

Defect triggers can be classified into six categories across three dimensions. Based on the quantifiability of the defect trigger, they can be divided into deviation-type defect triggers and non-deviation-type defect triggers (Chi et al. 2015). Deviation-type defect triggers are quantifiable trigger factors to quality defect, characterized by measurable parameters such as material nonconformities, environmental deviations, and measurement errors. Non-deviation-type defect triggers refer to trigger factors that are not directly characterized by specific quality deviation parameters, such as managerial deficiencies, inappropriate construction processes, human misjudgment, and complacent or lax managerial attitudes. Based on the objective formation process, they can be classified into defect triggers based on physical evolution mechanisms and defect triggers based on management evolution mechanisms (Lambers et al. 2024). Based on the relationship of their evolution, they can be categorized into defect triggers based on causal relationships and defect triggers based on correlational relationships (Aljassmi et al. 2016; Lee 2019).

This study analyzes the primary sources of quality non-conformance from the perspective of quality risk. The

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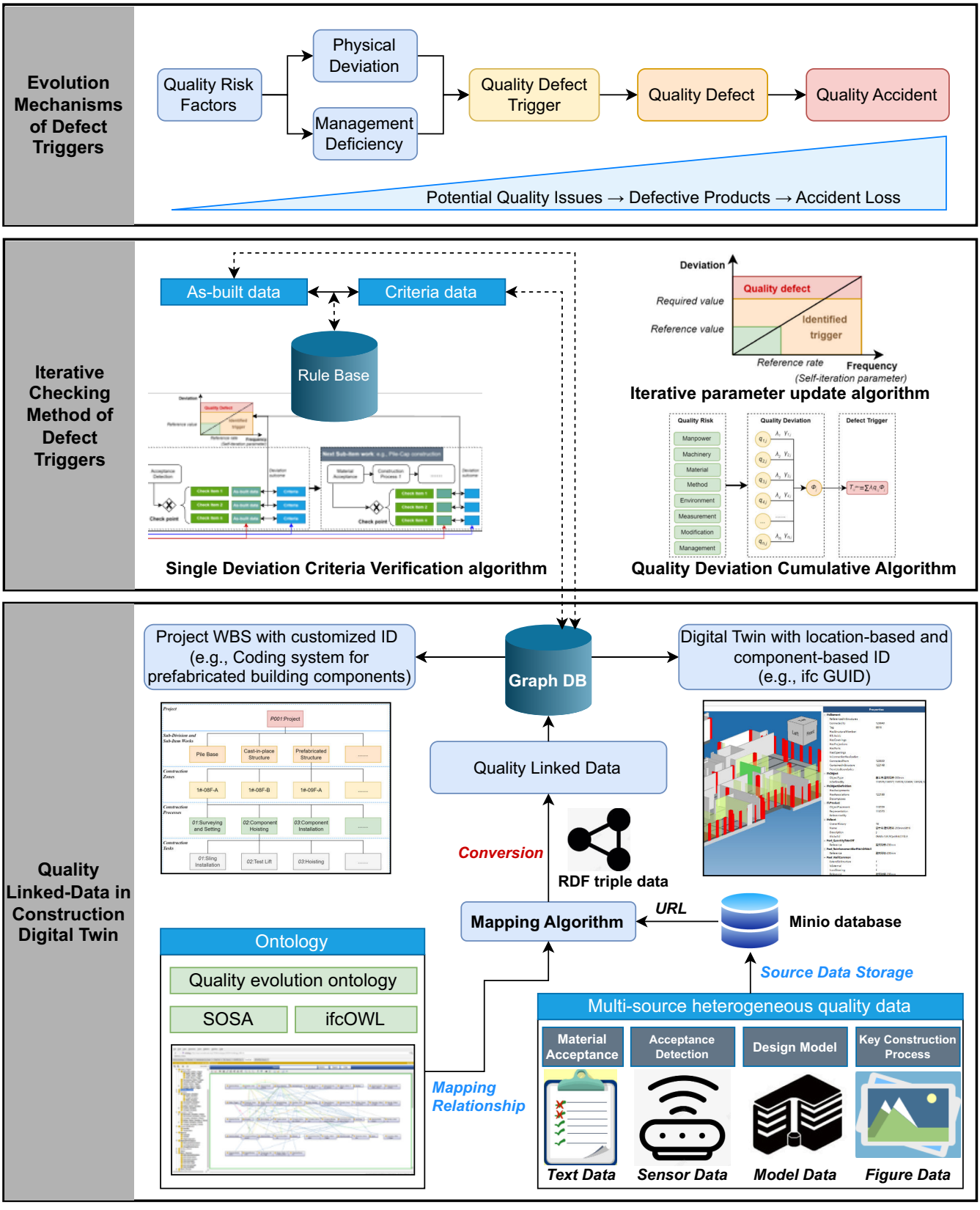


Fig. 3. Proposed framework for quality linked data development and defect trigger identification.

6M1E + modification" extension method is adopted, specifically analyzing eight sources: human, machinery, materials, methods, environment, measurements, modifications, and management (Gan et al. 2014; O'Connor and Koo 2020; Tarekegn Gurmu and Mahmood 2024). Quality defect triggers are embedded within quality deviations and noncompliant behaviors (Lambers et al. 2024; Lee 2019). They represent the formation processes of deviation-type defect triggers and non-deviation-type defect triggers. The evolutionary mechanism from construction quality deviations and noncompliant

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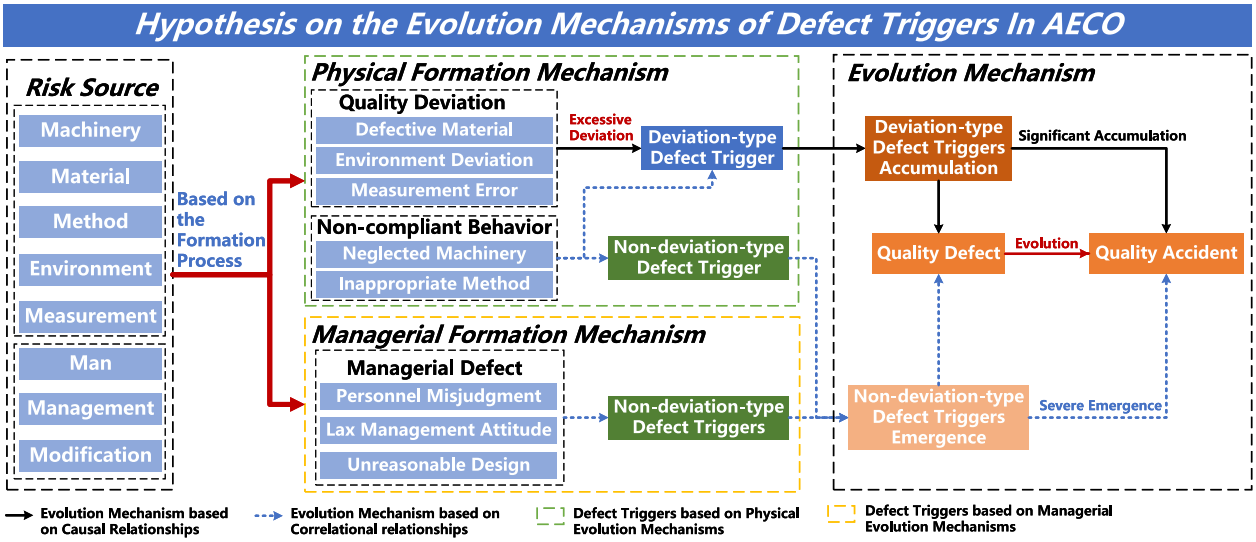


Fig. 4. Evolution mechanisms diagram of defect triggers.

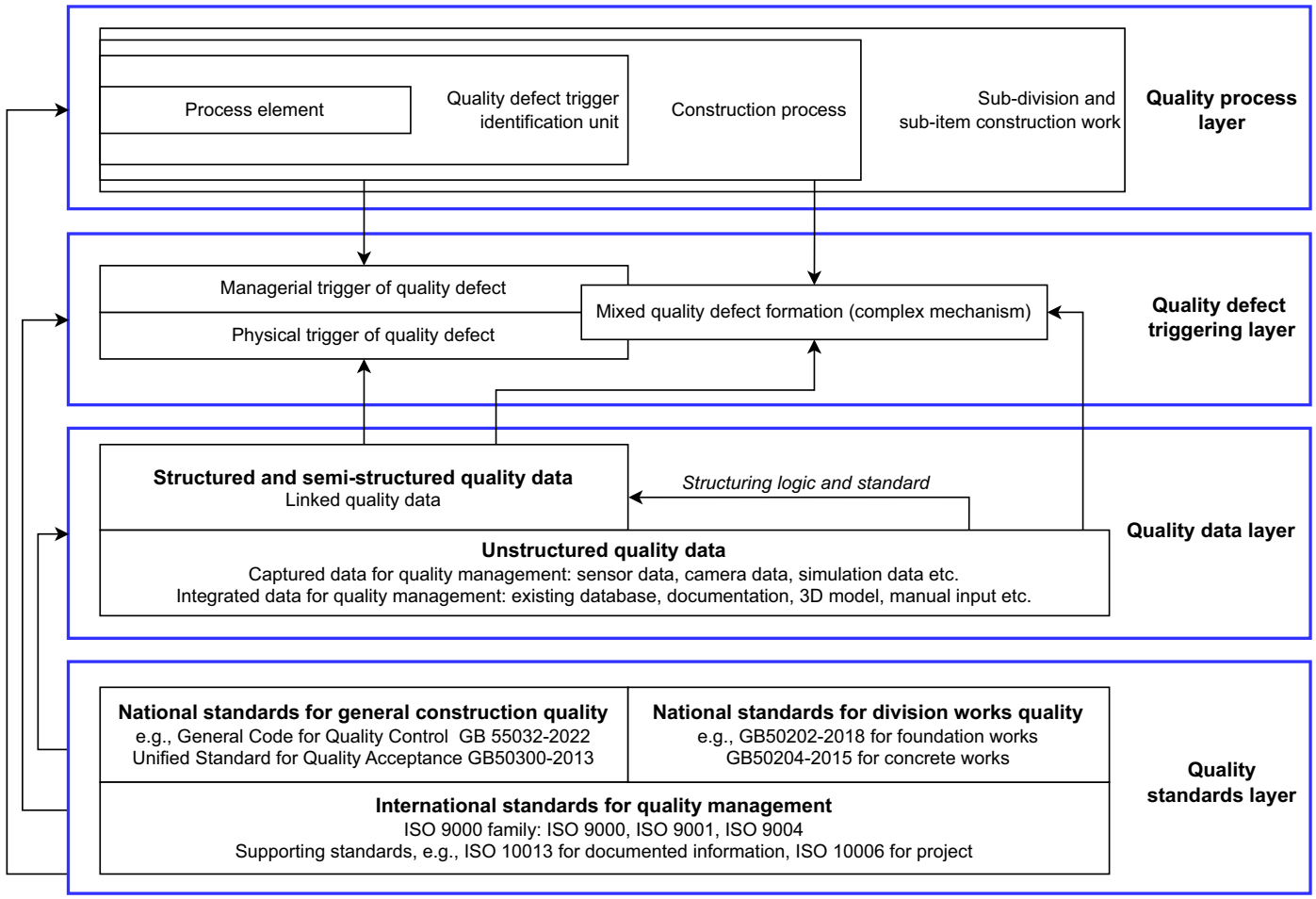


Fig. 5. Hierarchy of quality data and defect triggers.

behaviors to defect triggers, quality defects, and ultimately quality accidents is analyzed. The evolution mechanism hypothesis is presented in Fig. 4.

This study primarily focuses on the identification of deviation-type defect triggers. Quality deviations originate from eight categories of quality risk sources. As the magnitude of a deviation increases or its frequency becomes excessive, it may evolve into a quality defect trigger. This threshold-to-trigger interpretation accords with Statistical Process Control (SPC): observations exceeding Shewhart control limits for variables or the control limits of attribute charts (e.g., p-charts for defect proportions) constitute out-of-control conditions; moreover, memory-type procedures such

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as the cumulative sum (CUSUM) chart explicitly accumulate small departures from the target and are more efficient for detecting small sustained shifts (Diko et al. 2020). Specifically, a quality defect trigger is considered to occur under any of the following conditions:

- The deviation magnitude $q_{i,j}$, generated by quality risk source i in stage j exceeds a predefined threshold θ_q.
- The occurrence frequency $f_{i,j}$ exceeds a frequency threshold θ_f.
- The cumulative effect of minor deviations during a specific construction stage.

This relationship can be formalized as Eq. (1), where $\boldsymbol{I}_{(\cdot)}$ denotes the indicator function, which returns one if the condition holds and zero otherwise. The parameters λ_i, represent the attenuation coefficients associated with the contribution of each defect trigger to the final outcome, the values range of λ_i *is* within the [0,1]. Eq. (1) is aligned with risk-prioritization practice in Failure Modes and Effects Analysis (FMEA) and Risk Priority Number (RPN)–based ranking in quality management, whereby failure sources with greater severity, higher occurrence and poorer detectability are assigned higher priority (Liu et al. 2019).

The function ϕ_j, denotes the cumulative deviation function defined for stage j, as detailed in Eq. (2), where n_j refers to the source number of quality deviations present in stage j, and the parameter $\gamma_{i,j}$ reflects the degree to which the quality deviation $q_{i,j}$ influences the likelihood or severity of a specific quality defect trigger. This formulation is consistent with the Palmgren–Miner hypothesis of linear cumulative damage (Fatemi and Yang 1998).

Consistent with the foregoing definitions, Eqs. (1) and (2) are applied under the following conditions: the construction stage j is treated as a time interval with a stable data window; the deviation magnitude and occurrence frequency satisfy $q_{i,j} \geq 0$ and $f_{i,j} \geq 0$; and the number of sources present in stage j is n_j. The weighting parameters satisfy $\lambda_i \in [0,1]$ and $\gamma_{i,j} \geq 0$. The thresholds θ_q and θ_f are taken as SPC control limits. Measurement units are harmonized to industry practice [e.g., length in millimeters, temperature in Celsius, strength in megapascals, and relative humidity (RH) in %]. When λ_i values are elicited from experts, consistency and robustness checks are performed. In addition, the measurement system is required to exhibit adequate reproducibility, and the measurement uncertainty is assessed

$$T_{i,j}^{\text{dev}} = \lambda_i q_{i,j} \max[\boldsymbol{I}_{(q_{i,j} \geq \theta_q)}, \boldsymbol{I}_{(f_{i,j} \geq \theta_f)}] + \sum_{i=1}^{n_j} \lambda_i \cdot q_{i,j} \cdot \phi_j \quad (1)$$

$$\phi_j = \frac{\sum_{i=1}^{n_j} \gamma_{i,j} \cdot q_{i,j} \cdot \boldsymbol{I}_{(q_{i,j} < \theta_q)}}{\sum_{i=1}^{n_j} \gamma_{i,j} \cdot \theta_q} \quad (2)$$

Defect triggers have the potential to evolve into quality defects and even escalate into quality accidents if left unaddressed (Talebi et al. 2021; Zhong et al. 2025). Given that defect triggers may arise across different project stages and site-specific factors, this study employs a weighted scoring model and a defect trigger accumulation index (DTA) to quantitatively assess their potential impact. The relevant formulas are presented in Eqs. (3)–(5). In this framework, w_i denotes the weight coefficient associated with the quality defect trigger $T_{i,j}^{\text{dev}}$, DTA_j represents the cumulative defect trigger index for construction stage j, and $\text{DTA}_{\text{cum}}(N)$ denotes the overall accumulated defect trigger index across the entire project. A quality defect is considered to occur when the stage-level index DTA_j exceeds a predefined threshold θ_{def}; similarly, a quality accident is identified when the project-level index $\text{DTA}_{\text{cum}}(N)$ surpasses the accident threshold θ_{acc}. The formal definitions of these thresholds and trigger conditions are provided in Eqs. (6) and (7)

$$\text{DTA}_{i,j} = w_i \cdot T_{i,j}^{\text{dev}} \quad (3)$$

$$\text{DTA}_j = \sum_{i=1}^{n_j} \text{DTA}_{i,j} \quad (4)$$

$$\text{DTA}_{\text{cum}}(N) = \sum_{j=1}^{N} \text{DTA}_j \quad (5)$$

$$I_{\text{defect}} = \boldsymbol{I}\{\text{DTA}_j \geq \theta_{\text{def}}\} \quad (6)$$

$$I_{\text{accident}} = \boldsymbol{I}\{\text{DTA}_{\text{cum}}(N) \geq \theta_{\text{acc}}\} \quad (7)$$

The deviation-type defect trigger evolution mechanism is illustrated using the DHB-01 precast composite-slab. According to *Technical Standard for Assembled Buildings with Concrete Structure* [GBT 51231-2016 (MOHURD 2016)], the allowable production and installation deviation for length is $\theta_{\text{def}} = 5$ mm, and the length deviation defect trigger threshold and the position deviation defect trigger were set as $\theta_{q1} = \theta_{q2} = 0.9\,\theta_{\text{def}} = 4.5$ mm. For DHB-01, the design length is $L_{\text{des}} = 7{,}300$ mm and the fabricated length is $L_{\text{prod}} = 7{,}304.8$ mm, yielding $q_1 = L_{\text{prod}} - L_{\text{des}} = 4.8 \text{ mm} > \theta_{q1}$, so a deviation-type defect trigger arises.

Upon installation, the positional deviation along the slab's centerline is likewise $q_2 = 4.6 \text{ mm} > \theta_{q2}$, and another deviation-type defect trigger arises. Because both deviations occur in the same direction, the corresponding weight coefficients are set as $w_1 = w_2 = 1$. When these deviations are superimposed, the cumulative defect trigger index becomes: $\text{DTA}_j = \sum_{i=1}^{2} \text{DTA}_{i,j} = 1 \times 4.8 + 1 \times 4.6 = 9.4 > 5 = \theta_{\text{def}}$, thereby exceeding the permissible limit and constituting a quality defect. If this combined deviation is not corrected prior to concrete placement, the misalignment will inhibit proper integration of reinforcement and cast-in concrete, ultimately risking end-detachment of the slab—a quality accident.

Construction of Defect Trigger Identification Ontology

The identification of quality defect triggers requires mining and integrating heterogeneous, dispersed quality data from quality risk sources. To unify and semantically link these data, this study adopted a knowledge-graph solution: a domain ontology (TBox) that formalizes concepts, properties, and relations, and an instance layer (ABox) that populates them with project data for integration and reasoning (Abu-Salih and Alotaibi 2024). Within this solution, this study proposes a four-layer framework—quality standards, quality data, defect triggers, and construction processes. The standards layer specifies evaluation criteria [e.g., ISO 9000 (ISO 2015) and project codes]; the data layer aggregates sensor streams, BIM models, and inspection records in multiple formats; the defect trigger layer captures physical, managerial, and hybrid causes; and the process layer structures construction activities as the basic identification units. Linking these layers anchors triggers in their spatial, temporal, and logical contexts, enabling precise identification and prevention. Fig. 5 illustrates the hierarchy of quality data and defect triggers.

This study developed the TBox, named *Construction_DeTrig_Identification_Ontology*, which defines the semantic backbone for modeling quality defect trigger identification in

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construction processes. The *Construction_DeTrig_Identification_Ontology* was developed using Protégé (version 5.6.4) to support the identification of quality defect triggers in construction projects. The development process includes the following steps:

1. Domain specification: The scope is defined to focus on the quality defect trigger identification in construction.
2. Ontology reuse: Because defect trigger identification involves both criteria data acquisition and as-built data detection, the IfcOWL and sensor, observation, sample, and actuator (SOSA) ontologies were reused to represent design model data and detection actions.
3. Evolution-based extension: The domain-specific ontology *Construction_Quality_Evolution* is designed around the evolution of quality-defect triggers that originate from eight documented quality-risk sources. In this framework, as-built data—captured from field inspections, and criteria data—derived from design models or relevant specifications—are continuously compared to quantify inconsistencies along two orthogonal dimensions: (1) quality deviation, and (2) management defect. Significant discrepancies in the first dimension give rise to deviation-type defect triggers, whereas severe discrepancies in the second dimension produce non-deviation-type defect triggers; the progressive accumulation of either trigger type ultimately precipitates a quality defect. These causal relationships are explicitly encoded in the ontology *Construction_Quality_Evolution* via dedicated core classes and object

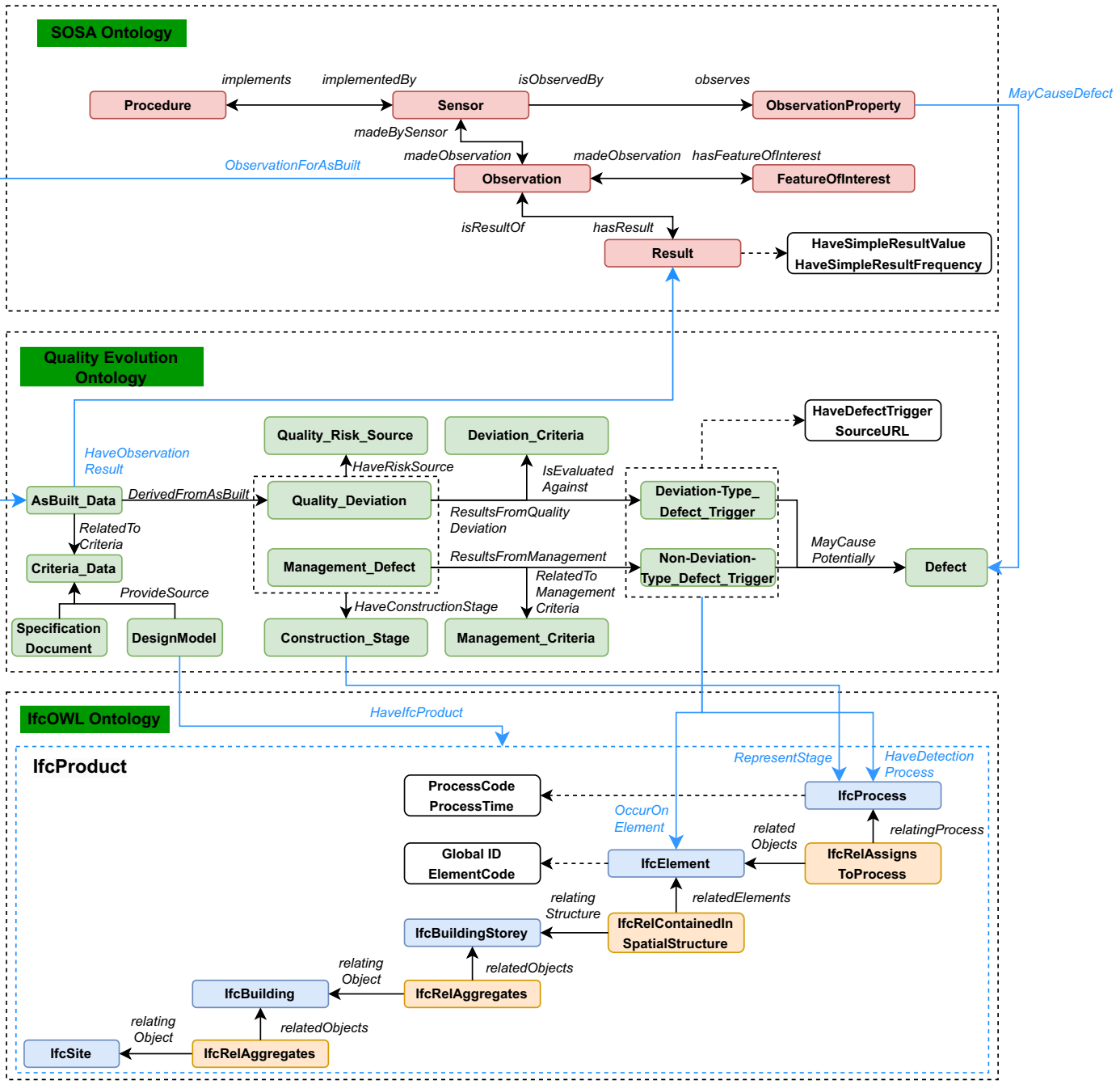


Fig. 6. Correlation between ontologies of *Construction_DeTrig_Identification_Ontology*.

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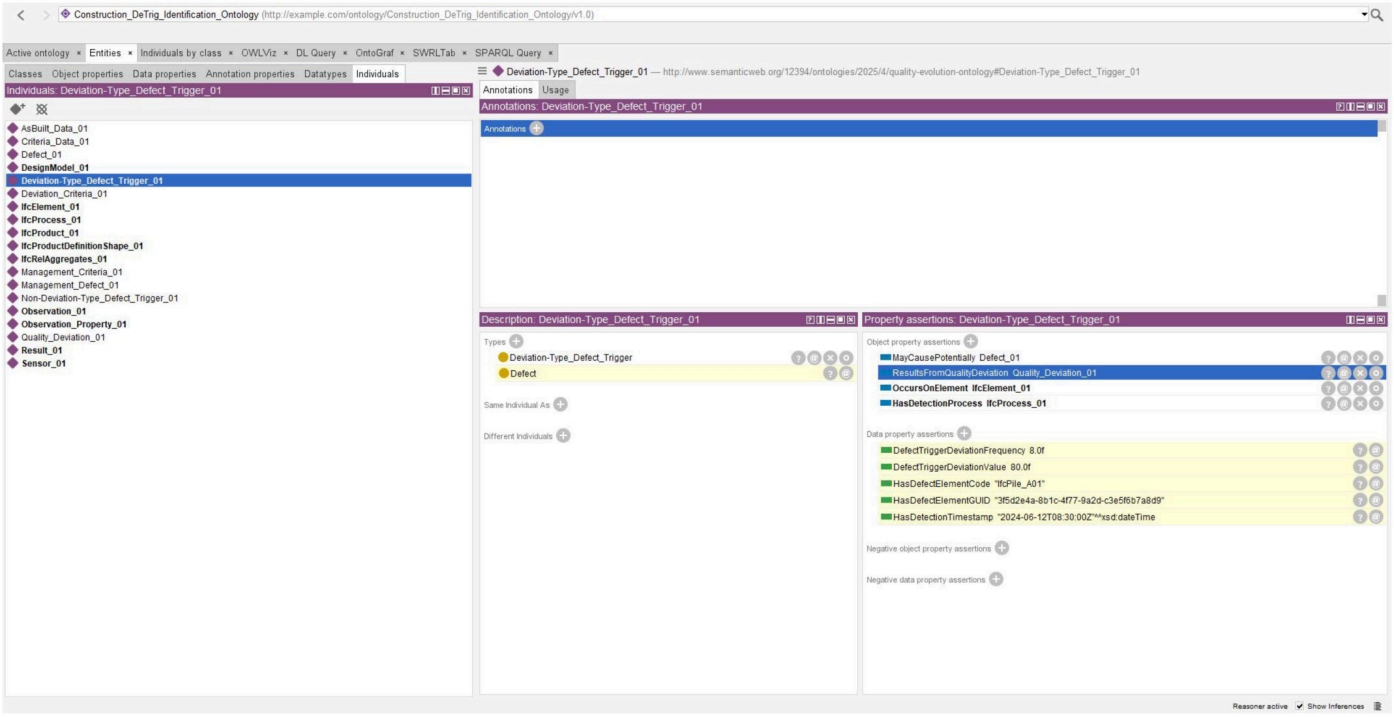


Fig. 7. Validation of logical consistency for *Construction_DeTrig_Identification_Ontology*.

properties, thereby providing a formal semantic foundation for automated quality-risk assessment and reasoning.

4. Ontology alignment: The *Construction_Quality_Evolution* ontology is aligned with the IfcOWL ontology (Pauwels and Terkaj 2016) and the SOSA ontology (Janowicz et al. 2019), establishing semantic interoperability among the three ontologies by a series of object properties.
5. Prototype verification: Representative individuals (e.g., Observation_01, Result_01, AsBuilt_Data_01, DesignModel_01, IfcProduct_01, IfcProductDefinitionShape_01, Criteria_Data_01, Quality_Deviation_01, Deviation_Criteria_01, and Deviation-Type_Defect_Trigger_01) were instantiated—using the reinforcement cage length of a cast-in-place pile as an example—to semantically link criteria data and as-built data. Semantic web rule language (SWRL) rules were then used to automatically retrieve and compare the relevant values. When a measurement exceeded the tolerance defined in Deviation_Criteria_01, the reasoner generated a Quality_Deviation_01. If the trigger conditions were satisfied, a Deviation-Type_Defect-Trigger_01 was activated. This process verified the ontology's functional logic, completeness, and consistency.

The *Construction_DeTrig_Identification_Ontology* structure and interrelations are illustrated in Fig. 6. The semantic web rule language (SWRL) reasoning results within the Protégé are illustrated in Fig. 7.

Quality Linked Data Development and Deviation Calculation

Ontology-based QLD is an effective means to integrate multisource heterogeneous quality data (Sheng et al. 2020). In this study, the *Construction_DeTrig_Identification_Ontology* serves as the TBox, and source files are mapped to ABox instances to realize unified semantics and linkage. From a data format perspective, the primary data types include IFC (.ifc extension), PDF (.pdf extension), CSV (.csv extension), and JPEG (.jpg extension). Along the temporal dimension, data are organized into four phases: design, production and transportation, on-site construction and installation, operation and maintenance. Along the spatial–procedural dimension, a top-down WBS/WP scheme decomposes the project into four levels: division works, construction zones, construction processes and specific tasks. Each quality record is tagged with a unique WBS code, enabling precise spatial positioning on site and temporal alignment with process progression. The WBS structure is shown in Fig. 8. The Hoisting task is coded P001-PrefabricatedStructure-1#-08F-A-02-03, and a prefabricated beam component is coded P001-PrefabricatedStructure-1#-08F-A-YZL-02-1, where YZL-02-1 denotes the first component of the second prefabricated beam type.

The core task is to align real-world engineering data with the TBox so that as-built and criteria data can be structurally populated in the ABox. Dedicated templates were developed to convert source data into RDF. As-built records (activities, results, equipment, indicators) are mapped to SOSA classes—Observation, Result, Sensor, and ObservableProperty. Design values and specification limitations become instances of criteria data. Four main formats (IFC, PDF, CSV, and JPEG) are parsed with format-specific scripts (e.g., Pandas + RDFLib, PyPDF2 + spaCy + RDFLib, IfcOpenShell, OpenCV + RDFLib) and transformed into RDF triples. The source files are stored in MinIO, and their URLs are embedded in the RDF for traceability. The mapping templates are illustrated in Fig. 9.

After completing the transformation of multisource heterogeneous quality data, the converted RDF files were imported into the GraphDB graph database along with the TBox *Construction_DeTrig_Identification_Ontology*. Separate named graphs were created to represent the ABox and TBox, denoted as the instances graph and the ontology graph, respectively. SPARQL queries were implemented to retrieve the ABox quality data and traverse the relational logic defined by the TBox. This approach enabled automated calculation of quality deviations by comparing as-built data and criteria data within the unified semantic framework. The pseudocode illustrating the SPARQL queries used for this process is provided in Algorithm 1.

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Algorithm 1. Decision Logic for Quality Deviation Computation in SPARQL Queries

Quality Deviations Computation

Input: *criteriaGraph* ← **GRAPH** ‹*…/criteriainstances*›;
asbuiltGraph ← **GRAPH** ‹*…/asbuiltinstances*›;
Output: *outputGraph* ← **GRAPH** ‹*…/deviationinstances*›;

```
CLEAR outputGraph
FOR each critInst IN criteriaGraph WHERE rdf:type = qeo:Criteria_Data:
    critVal ← getLiteral(critInst,qeo:CriteriaValue)
    asbInst ← IRI(replace(str(critInst), "Criteria_Data", "AsBuilt_Data"))
    asbVal ← getLiteral(asbInst IN asbuiltGraph, qeo:AsBuiltValue)
    deviation ← toDouble(asbVal) - toDouble(critVal)
    nameTail ← substringAfter(str(critInst),"Criteria_Data_")
    devInst ← IRI(concat(outputGraph.baseIRI, "Quality_Deviation_",encode(nameTail)))
    devLabel ← concat("QualityDeviation",nameTail)
    INSERT INTO outputGraph:
        (devInst,rdf:type,    qeo:Quality_Deviation)
        (devInst,cdti:DerivedFromAsBuilt,asbInst)
        (devInst,cdti:DerivedFromCriteria,critInst)
        (devInst,qeo:ConstructionDeviationValue,deviation)
        (devInst,rdfs:label,    devLabel)
END FOR
```

Defect Trigger Identification Based on Single Deviation Criteria Verification

To achieve comprehensive and accurate identification of quality defect triggers, the defect trigger identification workflow should be established within the fundamental construction elements of building projects. According to the *Unified Standard for Construction Quality Acceptance of Building Engineerings* [GB 50300-2013 (MOHURD 2013)], construction projects are hierarchically divided into division works, subdivision works, and subitem works, with subitems representing the most basic construction processes. Quality inspection is conducted at predefined checkpoints and includes material acceptance, key construction process inspections (covering concealed works, installation work, and measurement work), and subitem work acceptance.

Each checkpoint involves extracting as-built data and criteria data from QLD to calculate deviations, whose magnitude and frequency are analyzed to determine whether they constitute defect triggers. Once all subitem works are completed and accepted, the entire division work undergoes spot-check acceptance inspections. Accordingly, this study proposes that the fundamental unit for defect trigger identification is the acquisition and comparison of QLD at key inspection points, supporting a quality deviation-based approach for automated defect trigger identification. The proposed workflow is illustrated in Fig. 10.

According to the evolution mechanisms of defect triggers, a quality deviation may constitute a defect trigger when its magnitude closely approaches the tolerance limits defined by relevant standards or when its frequency becomes abnormally high. To support identification, three key indicators are proposed: the required value, the reference value, and the reference rate. Specifically, a quality deviation exceeding the required value is classified as a quality defect. When the deviation exceeds the reference value but remains within the required value, it is considered a defect trigger. Additionally, if the frequency of a deviation surpasses the reference rate, it may also be identified as a defect trigger due to its cumulative impact. The identification criteria for quality defect triggers are illustrated in Fig. 11.

Current AECO industry standards lack a systematic approach to reference value. To address this gap, this study introduces two reference thresholds, drawing on critical alerting concepts from environmental monitoring and industrial quality control domains:

1. Reference value: The reference value is established at the 90th percentile of historical deviation distributions. This threshold ensures that only the most extreme 10% of observed deviations are classified as potential defect triggers. The 90th percentile of historical deviation distributions has been widely applied in SPC, nonparametric SPC, and water quality monitoring (Lake et al. 2002; Qiu 2018; Saghir and Kosztyán 2023). The FHWA source has been verified as Burati et al. (2003), optimal procedures for quality assurance specifications, FHWA-RD-02-095, full payment is made only when the percent within limits (PWL) quality level reaches 90 (Bennett and White 2025; Burati et al. 2003; Federal Aviation Administration 2018).
2. Reference rate: Based on the Pareto principle (Durdyev and Ismail 2012), a 20% occurrence threshold is set for identifying systemic risk. Based on statistical analyses of AECO nonconformance (NCR) and quality defect records, respectively, Love et al. (2018) and Del Solar et al. (2016) reported that a minority of quality events or construction stages leads to most defect occurrences, approximating the Pareto principle. Accordingly, this study adopts the Pareto principle to determine the reference rate. If the frequency of similar quality deviations in a specific process or inspection batch exceeds 20%, the condition is flagged as a potential defect trigger.

Both the reference value and reference rate should be derived from industry statistical data, then periodically updated and

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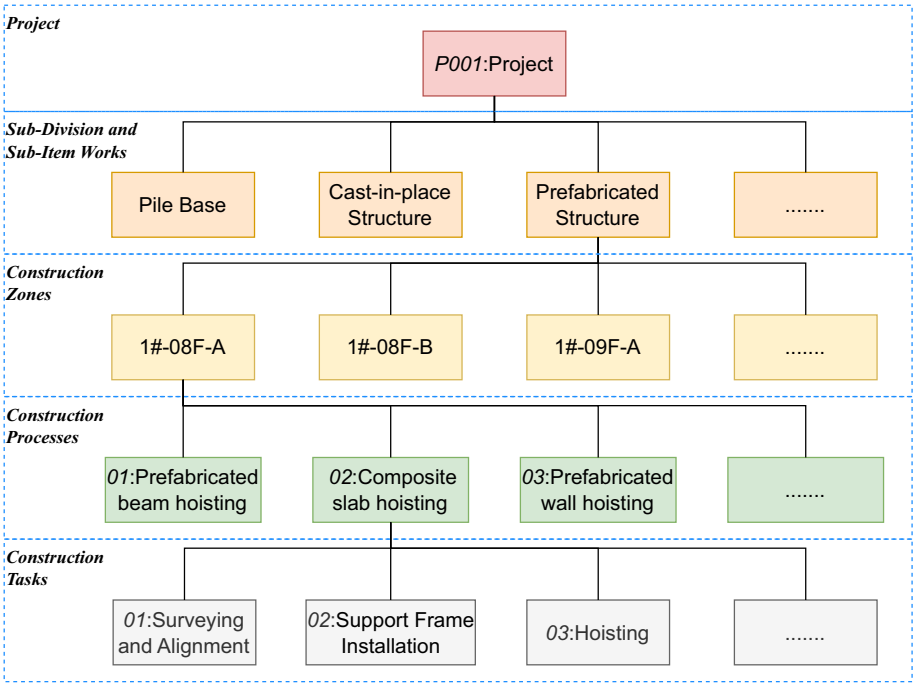


Fig. 8. Four-level WBS hierarchy for construction project.

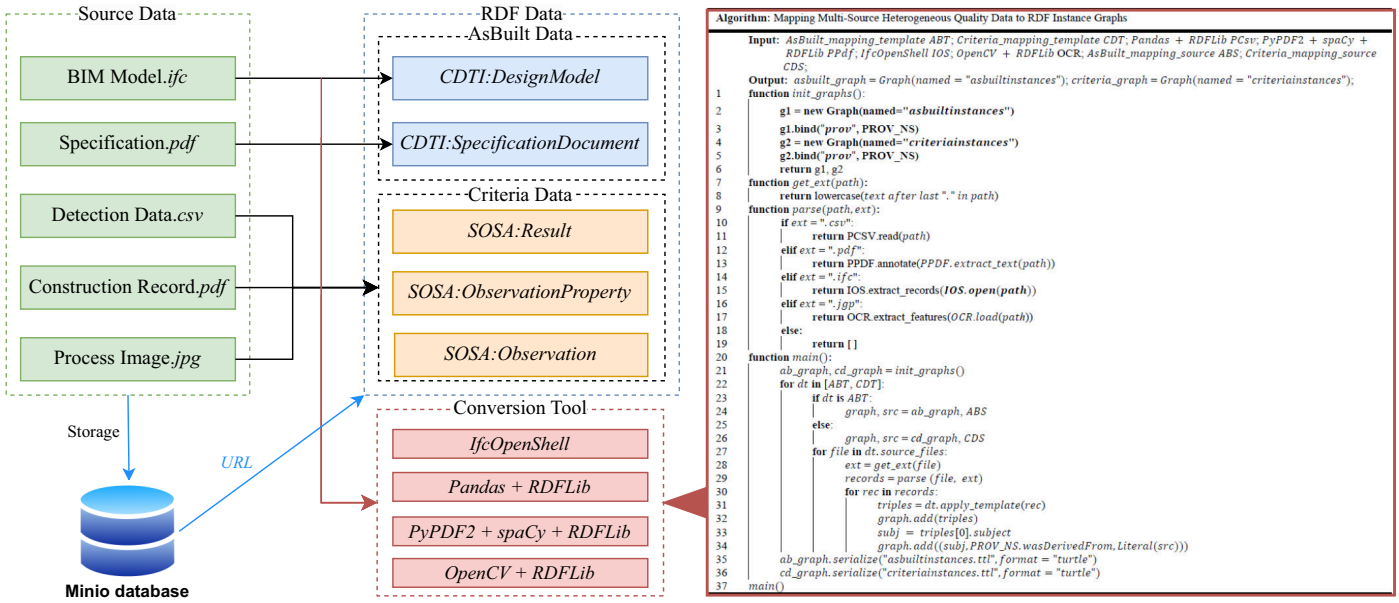


Fig. 9. Mapping templates between source data and RDF data. (Adapted from Ye et al. 2024.)

calibrated using ongoing deviation data and expert validation. This process is designed to maintain an optimal balance between trigger sensitivity and detection reliability over the project life cycle.

The study conducts rule-based verification based on the reference value and reference rate thresholds to enable the identification of individual defect triggers. The SPARQL query functionality of the GraphDB graph database was employed to develop query pseudocode supporting defect trigger detection. The pseudocode covers material acceptance, key construction process inspections, acceptance detection, and division work acceptance. The corresponding SPARQL pseudocode is presented in Algorithm 2.

Defect Trigger Identification Based on Quality Deviation Cumulative Model

During the construction phase, when the magnitude of an individual deviation $q_{i,j}$ remains below the single-event threshold θ_q, traditional threshold-based assessment methods typically classify it as acceptable and disregard its subsequent impact. However, practical experience demonstrates that the accumulation of numerous minor deviations within the same phase can likewise evolve into significant quality hazards. This accumulation effect is explicitly represented by the cumulative term in Eq. (1): $\sum_{i=1}^{n_j} \lambda_i \cdot q_{i,j} \cdot \phi_j$. Here, λ_i denotes the attenuation coefficients, which reflect the

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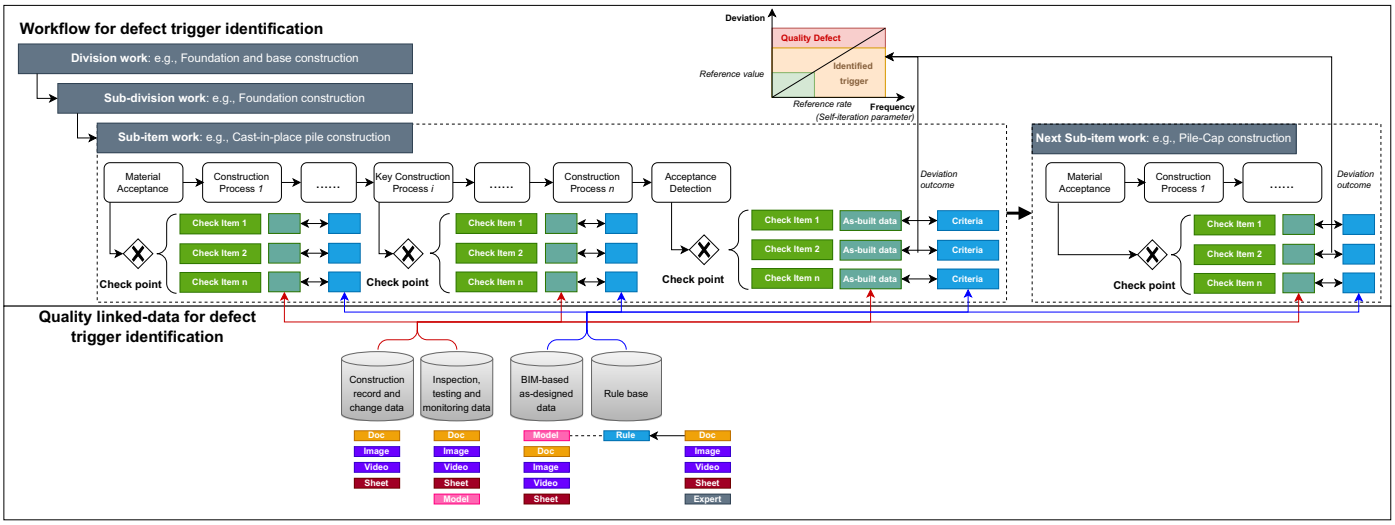


Fig. 10. Workflow for quality deviation calculation and defect trigger identification.

proportional contribution of deviations from risk source i to the formation of a defect trigger. These coefficients are determined through historical data fitting or expert assessment.

The parameter ϕ_j represents the stage accumulation saturation, quantifying the extent to which the allowable deviation quota in stage j has been consumed. The detailed formulation is provided in Eq. (2). As this value increases, the likelihood of triggering a defect correspondingly rises. In Eq. (2), $\gamma_{i,j}$ denotes the sensitivity coefficient, which represents the relative sensitivity or importance of deviations from risk source i in stage j with respect to the formation of overall quality defect triggers (Fig. 12).

To enable the consideration of the quality deviation cumulative effect in practical engineering applications, the parameters λ_i, ϕ_j, $\gamma_{i,j}$ were defined as data properties named HaveLambda, HavePhi, and HaveGamma during the development of the *Construction_DeTrig_Identification_Ontology*. These properties were assigned to the classes QualityRiskSource, IfcProcess, and QualityDeviation. After importing the TBox into the GraphDB repository, the corresponding ABox instances were populated with empirical data for λ_i and $\gamma_{i,j}$, which were mapped to their designated data properties. Subsequently, SPARQL queries were executed to calculate the corresponding values of ϕ_j. Through this process, Eq. (1) was applied to compute $T_{i,j}^{\text{dev}}$, thereby supporting the automated identification of defect triggers.

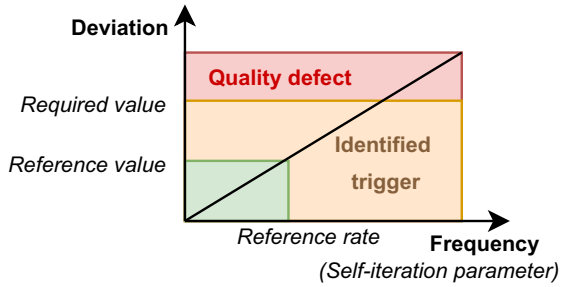


Fig. 11. Identification criteria of quality defect triggers.

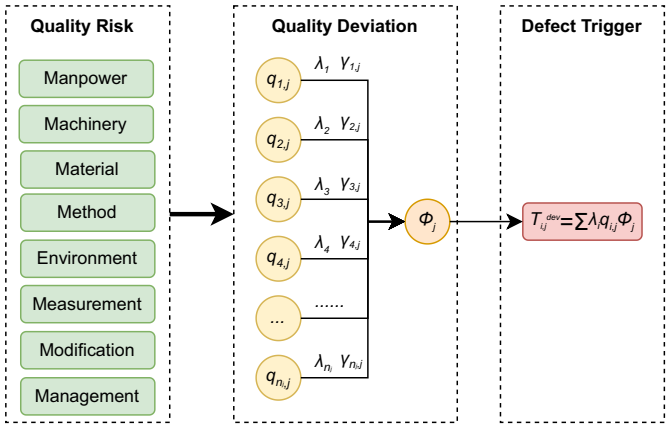


Fig. 12. Quality deviation cumulative model.

Implementation and Validation

To assess the applicability of the proposed framework for quality defect trigger identification in real-world construction projects, this study selects a practical construction project case for validation. The implementation includes testing the framework prototype (which encompasses the mapping of multisource heterogeneous quality data and defect trigger identification based on a graph database). Additionally, the framework and prototype were evaluated based on the feedback from AECO domain experts. The specific process is illustrated in Fig. 13.

Use Case Introduction

The use case is a prefabricated construction project in China, located in the Nanshan District of Shenzhen, China. This project is a 46-story high-rise affordable housing development with a typical cross-shaped architectural layout. The total building area is approximately 46,000 m^2, including the central vertical transport area and four residential zones. The project employs prefabricated construction technology, with the main structural framework constructed using traditional cast-in-place methods, whereas the primary enclosing structures, such as floors and external walls, use prefabricated components. The building incorporates prefabricated composite-slabs, external walls, and protruding window panels.

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Algorithm 2. Decision Logic for Identifying Defect Triggers

Identification of Quality Defect Triggers

Input: ***mad***: *material_acceptance_deviation*; ***maf***: *material_acceptance_frequency*;
madr: *material_acceptance_deviation_reference*; ***mafr***: *material_acceptance_frequency_reference*;
cdd: *construction_detection_deviation*; ***cdf***: *construction_detection_frequency*;
cddr: *construction_detection_deviation_reference*; ***cdfr***: *construction_detection_frequency_reference*;
ced: *construction_environment_deviation*; ***cef***: *construction_environment_frequency*;
cedr: *construction_environment_deviation_reference*; ***cefr***: *construction_environment_frequency_reference*;
wad: *work_acceptance_deviation*; ***waf***: *work_acceptance_frequency*;
wadr: *work_acceptance_deviation_reference*; ***wafr***: *work_acceptance_frequency_reference*;

Output: *DeTi*: defect_triggers

```
Initialize: DeTi
for i∈[1,n] do
    division_work_i
    for j∈[1,m] do
        sub_item_work_j
        material_acceptance_j
        if mad ≥ madr then
            DeTi.add(material_acceptance_j)
        else if maf ≥ mafr then
            DeTi.add(material_acceptance_j)
        end
        for k ∈ [1,l] do
            key_construction_process_k
            construction_detection_k
            if cdd ≥ cddr then
                DeTi.add(construction_detection_k)
            else if cdf ≥ cdfr then
                DeTi.add(construction_detection_k)
            end
            environment_monitor_k
            if ced ≥ cedr then
                DeTi.add(environment_monitor_k)
            else if cef ≥ cefr then
                DeTi.add(environment_monitor_k)
            end
        end
        sub_item_work_acceptance_j
        if wad ≥ wadr then
            DeTi.add(sub_item_work_acceptance_j)
        else if waf ≥ wafr then
            DeTi.add(sub_item_work_acceptance_j)
        end
    end
    division_work_acceptance_i
    if wad ≥ wadr then
        DeTi.add(division_work_acceptance_i)
    else if waf ≥ wafr then
        DeTi.add(division_work_acceptance_i)
    end
end
return DeTi
```

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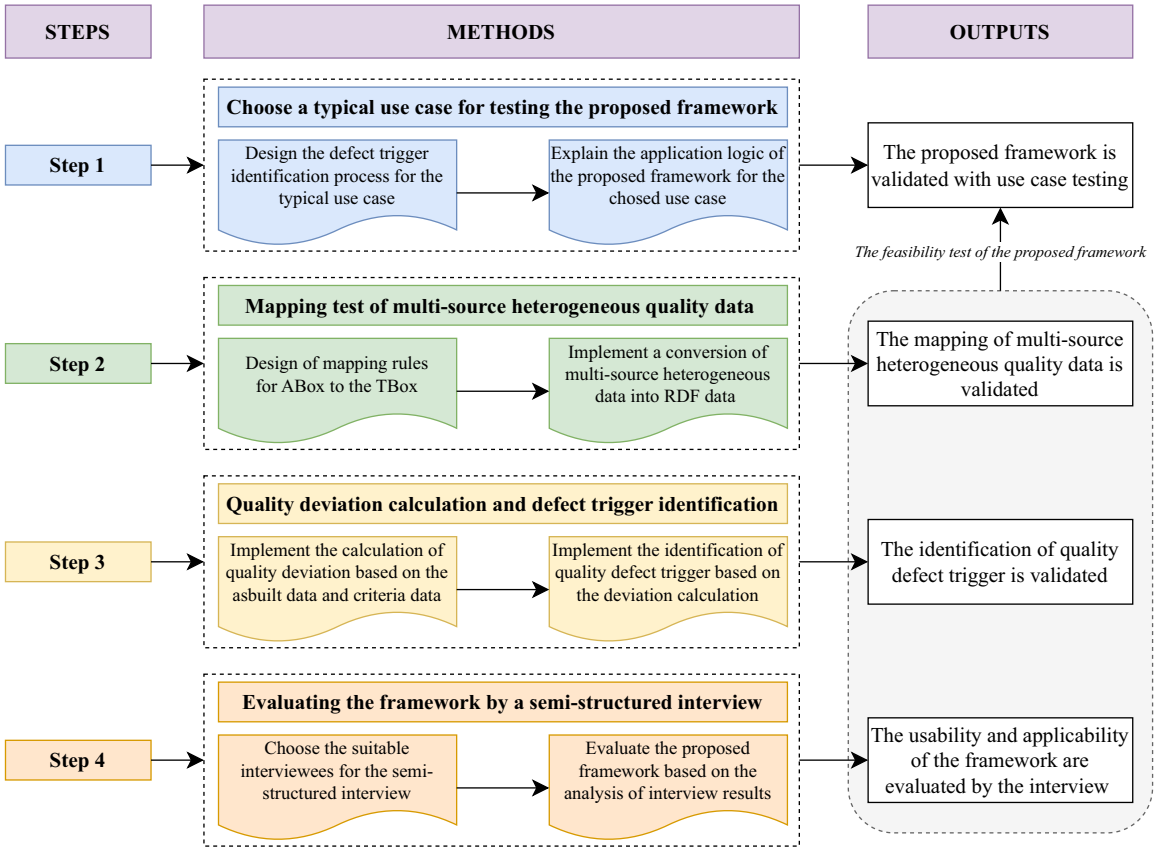


Fig. 13. Implementation and validation steps. (Adapted from Ye et al. 2024.)

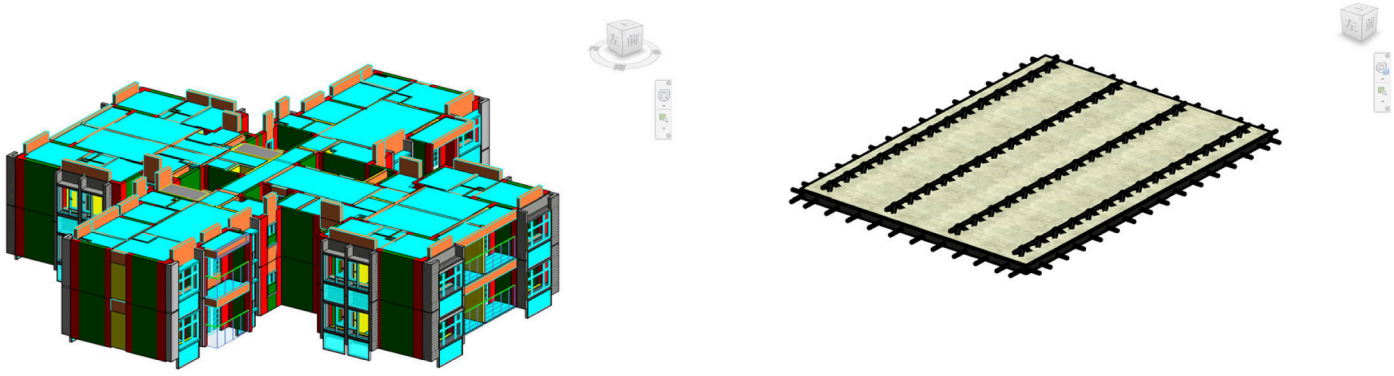


Fig. 14. BIM model of the standard floors and the prefabricated composite-slab component.

BIM technology was utilized for design work, particularly for the detailed design required for prefabricated construction. The prefabricated components for the project are produced in a factory located 140 km from the construction site, with transportation via truck. Upon on-site acceptance, the components are hoisted and installed. The construction sequence follows the order of (1) on-site casting of core structures and columns, (2) installation of prefabricated composite-slabs, and (3) installation of prefabricated external walls and protruding window panels. Considering that prefabricated project involves more stages compared with traditional cast-in-place construction, the project management team established and enforced a more rigorous quality management framework. Fig. 14 presents the BIM model of the standard floors and the prefabricated composite-slab component used in the case study.

Implementation

Defect Trigger Identification Workflow for Prefabricated Composite-Slab Installation

This study uses the prefabricated composite-slab installation procedures as a case example. Following the generic quality defect trigger identification workflow shown in Fig. 10, a process-specific identification workflow was developed for prefabricated composite-slab installation. The detailed installation workflow and its six check points are illustrated in Fig. 15, comprising: incoming inspection of prefabricated composite-slabs, on-site recheck of surveying, elevation adjustment of support frames, positional verification during slab hoisting, detection of surface

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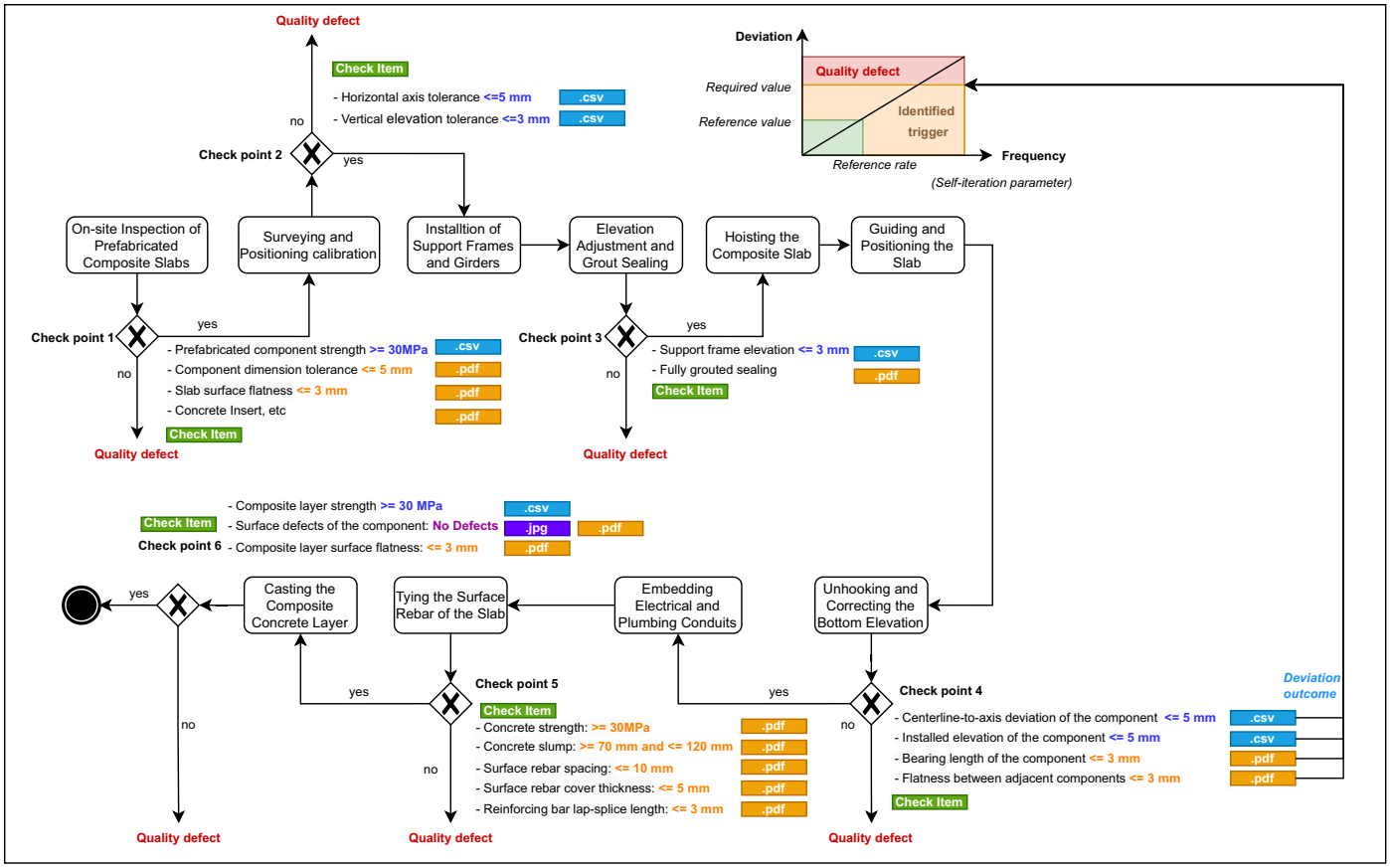


Fig. 15. Workflow of defect trigger identification in prefabricated composite-slab installation procedures.

rebar tying on the slab, acceptance testing after casting the composite topping layer.

The specific inspection items associated with each checkpoint are enumerated in Fig. 15. The project adopts a WBS-based numbering system, where the element code of the prefabricated composite-slab is P001-PrefabricatedStructure-1#-11F-A-DHB-07-1, and the numbering for the Elevation Adjustment and Grout Sealing construction task is P001-PrefabricatedStructure-1#-11F-A-02-04.

Multisource Heterogeneous Quality Data Storage and RDF Transformation

The study first collected multisource heterogeneous quality data generated during the installation of the prefabricated composite-slabs, covering design data, process record data, and inspection data. The design data includes the 13. Prefabricated Component Standard Layer Model (an IFC-format BIM model) and the 14. Design Document (a PDF-format design specification), which were used to define the design requirements and regulatory requirements of the components. The process record data primarily consisted of manually record quality acceptance documents, such as the 2. Incoming Component Quality Acceptance Report, which documented inspection results for quality indicators such as dimensions and flatness of incoming prefabricated components.

The inspection data were obtained using professional measurement devices, including total stations and concrete rebound hammers. The inspection data were stored in CSV format to reflect actual measurements taken on the construction site. All of these multisource heterogeneous data were stored in MinIO, an S3-compatible object storage system, with each file assigned a URL accessible by the processing scripts, facilitating a unified distributed storage and retrieval system. The data acquisition and storage process is illustrated in Fig. 16(a).

After completing the storage of multisource heterogeneous quality data, the study developed data conversion scripts in Python, implemented in the Anaconda environment using the Spyder IDE, based on the mapping templates shown in Fig. 9. These scripts enabled automated data integration and semantic representation. Fig. 16(b) uses the flatness and overlapping length inspection data of the prefabricated composite-slabs as an example to demonstrate the core functions of the script: (1) reading the source files stored in MinIO via URLs; (2) extracting key data from PDF files and parsing unstructured textual information into structured entries; and (3) transforming the extracted data into RDF triples compliant with the semantic data model, in accordance with the definitions in the TBox (ontology terminology layer), thereby completing the standardized mapping of heterogeneous data into the RDF graph data.

Knowledge Graph–Based Workflow for Defect Trigger Identification in Prefabricated Composite-Slab Installation

After converting the multisource heterogeneous quality data into RDF graph data, both the RDF files (ABox) and the *Construction_DeTrig_Identification_Ontology* (TBox) were imported into the GraphDB graph database to perform quality deviation calculation and defect trigger identification. The procedure, illustrated in Fig. 17, consists of five steps:

1. Create the RelatedToCriteria object property: After importing the As-Built_Data and Criteria_Data instances, the

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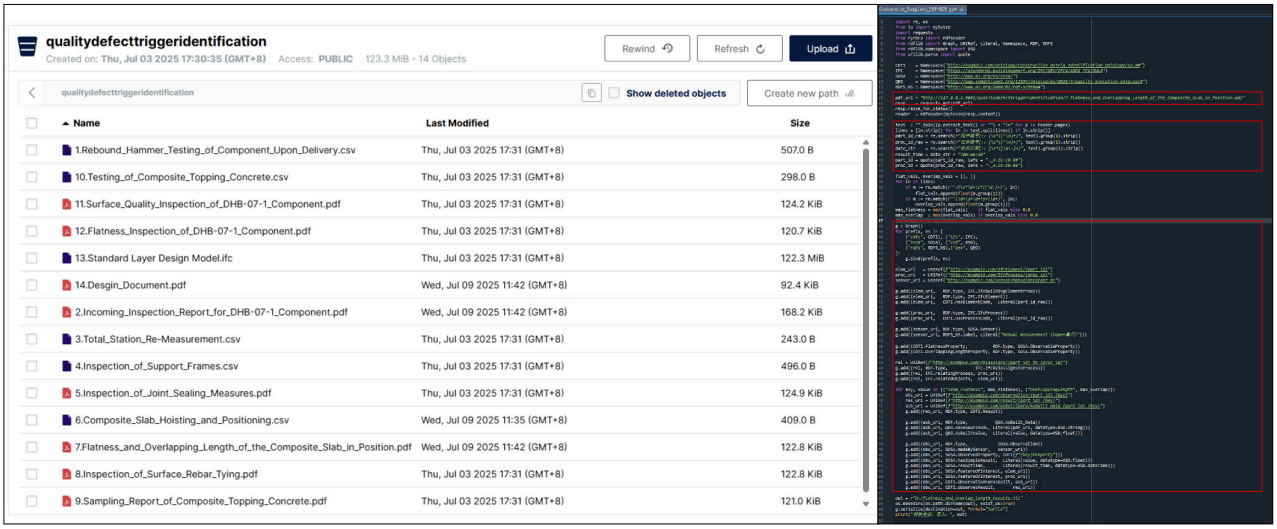


Fig. 16. Storage and conversion of multisource heterogeneous quality data for prefabricated composite-slab.

TBox object property RelatedToCriteria was invoked to link each As-Built_Data instance to its corresponding Criteria_Data instance.

2. Quality deviation calculation: The TBox object property DerivedFromAsBuilt (between Quality Deviation and As-Built Data) was used to generate Quality Deviation instances. QualityDeviationValue was computed as the difference between the paired AsBuiltValue and Criteria-value, and QualityDeviationFrequency was calculated as the ratio of the occurrence count of this deviation type to the total number of deviations in the project. The SPARQL queries implementing these calculations are shown in Fig. 17(a).
3. Create the IsEvaluatedAgainst object property: Using the TBox object property IsEvaluatedAgainst (linking Quality Deviation to Deviation Criteria), project data were used to populate Deviation Criteria with ToleranceDeviationValue and ToleranceDeviationFrequency. Corresponding links between Quality Deviation and Deviation Criteria instances were then established.
4. Defect trigger identification: Quality_Deviation instances whose deviation magnitude or frequency exceeded the associated Deviation_Criteria thresholds were classified as Defect Triggers, and corresponding Defect_Trigger instances were created. The deviation magnitude and frequency were written to the data properties DefectTriggerDeviationValue and DefectTriggerDeviationFrequency. In the installation case for the prefabricated composite-slab coded P001-PrefabricatedStructure-1#-11F-A-DHB-07-1, two deviation-type quality defect triggers were identified: an x-axis alignment deviation and the seam flatness between adjacent slabs. Their DefectTriggerDeviationValue values were 4.63 and 2.9 mm, both exceeding the corresponding ToleranceDeviationValue thresholds of 4.5 and 2.7 mm.
5. Visualization of identified defect triggers: Because the *Construction_DeTrig_Identification_Ontology* contains complex inter-class relationships, an Overview graph was generated for intuitive visualization. By invoking the object properties among TBox core classes and loading all instance data, the Overview graph was construct. The Overview graph, shown in Fig. 17(b), displays the relationships among the principal classes and the identified defect triggers. Fig. 17(b) illustrates the seam flatness quality defect trigger, with the HaveDefectTriggerSourceURL data property used to record the URL of the PDF source file for this defect trigger.

Quantitative Evaluation and Comparative Analysis

To quantitatively verify the efficiency of the proposed quality defect trigger identification method, this study solidifies the defect trigger identification of prefabricated composite-slab into reproducible test cases. Three new defect triggers, namely support frame elevation error, reinforcement lap length deviation of composite-slab, and composite-slab length deviation, were manually added to the original multisource heterogeneous quality data. Test cases 1, 2, and 3 were developed, each incorporating an increasing number of defect triggers, with three, four, and five triggers respectively. Ten participants were invited for the experiment. Five were industry practitioners with experience in quality management (IDs EX1–EX5). Five were civil engineering graduate students without such experience (IDs FR1–FR5). Their basic information is summarized in Table 1.

Quality defect triggers in the three test cases were identified by participants using the Risk Checklist of Quality Defect Trigger. The proposed method was applied to the same tasks for comparison. Performance was assessed using three metrics: deviation calculation accuracy, defect trigger identification accuracy, and task completion time. The checklist used in the tests is provided in Table S1.

The summary of the comparative experiment is presented in Table 2. For deviation calculation accuracy, experienced participants achieved 86%–98%, whereas inexperienced participants achieved 76%–88%. For defect trigger identification accuracy, the former achieved 81%–100%, whereas the latter achieved 48%–85%. For task completion time, the former required 1,171–1,307 s, whereas the latter required 1,576–1,662 s. A pronounced learning effect was observed in both groups as Cases 1–3 were completed, with task time decreasing progressively. By contrast, the proposed method outperformed on all metrics (ID: PRO). Its defect trigger identification accuracy reached 100%, and its task average time was 371 s, faster than the traditional manual process.

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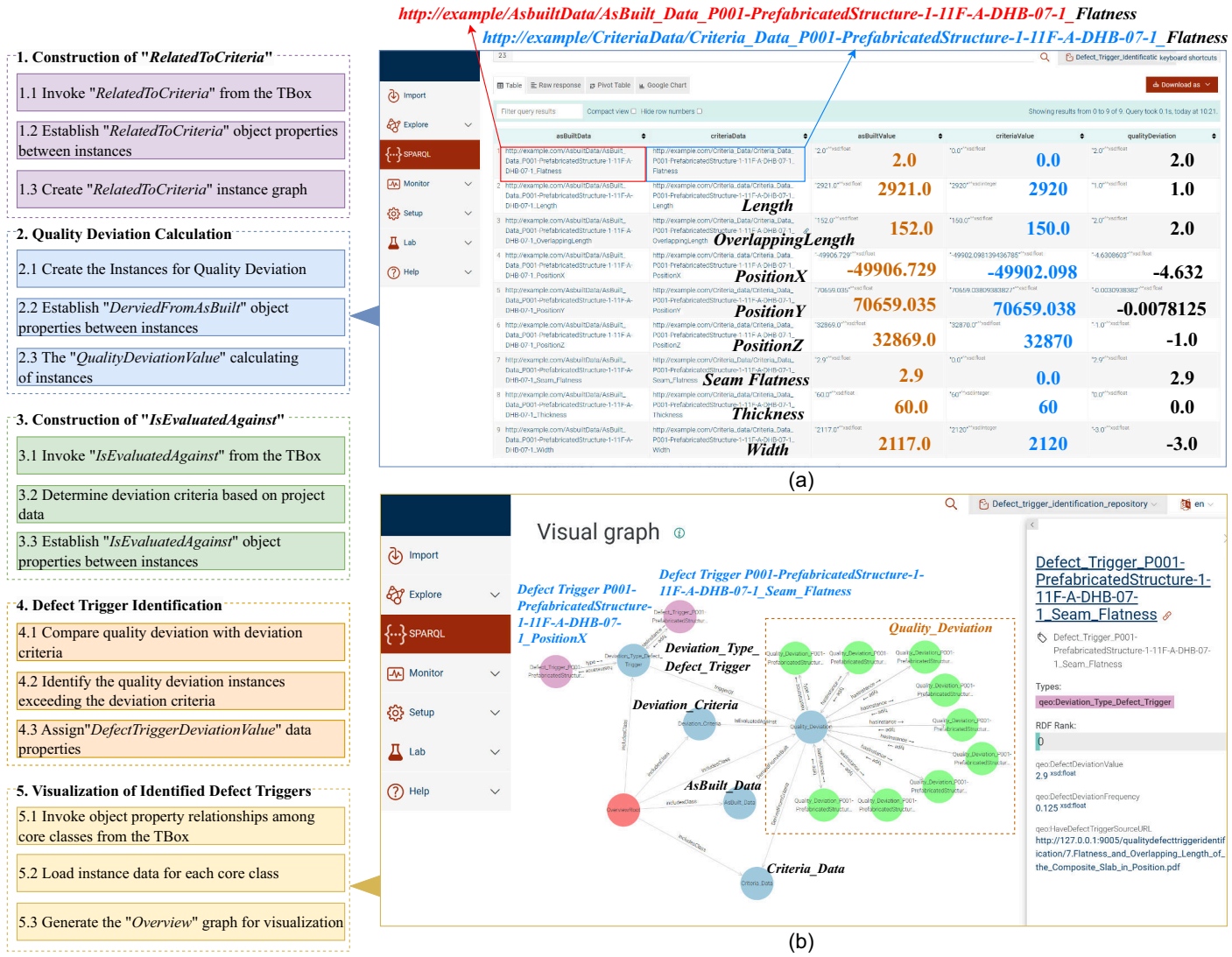


Fig. 17. Quality deviation calculation and defect trigger identification based on GraphDB.

Table 1. Basic information of 10 comparative experiment participants

ID	Organization	Position	Working/study experience	BIM experience
EX1	Owner	Project manager	Over 15 years	No
EX2	Supervision agency	Supervision engineer	Over 15 years	Yes
EX3	Main contractor	Quality manager	5–10 years	Yes
EX4	Main contractor	Project engineer	Below 5 years	Yes
EX5	University	Postdoctoral fellow	Below 5 years	Yes
FR1	University	Master's student	1st year	No
FR2	University	Master's student	2nd year	Yes
FR3	University	Master's student	2nd year	No
FR4	University	Ph.D. student	2nd year	Yes
FR5	University	Ph.D. student	4th year	Yes

Functional Interface Design of the Quality Defect Trigger Identification System

To address future application needs, the front end of the defect trigger identification system was developed using the Modao app. The interface is organized into five core functional regions: Function A, Data Upload, which ingests multisource heterogeneous quality data files together with standard or rule files; Function B, Rule Input, enabling manual specification of defect trigger identification rules; Function C, Data Storage and RDF Transformation, which supports querying of uploaded sources and execution of RDF triple conversion; Function D, Quality Deviation Calculation Results, providing execution of deviation computations and querying of computed outputs; and Function E, Defect Trigger Identification Results and Visualization, which performs the identification task

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Table 2. Result of 10 comparative experiment participants

ID	Accuracy of deviation calculation				Accuracy of defect trigger identification				Total time (s)			
	1 (%)	2 (%)	3 (%)	Average (%)	1 (%)	2 (%)	3 (%)	Average (%)	1	2	3	Average
FR1	71	79	86	79	33	50	60	48	2,137	1,464	1,127	1,576
FR2	64	86	100	83	33	75	100	69	2,223	1,615	1,147	1,662
FR3	79	86	93	86	67	75	100	81	2,263	1,565	1,128	1,652
FR4	86	86	93	88	100	75	80	85	2,186	1,671	1,024	1,627
FR5	57	86	86	76	33	75	80	63	2,176	1,501	1,075	1,584
EX1	79	86	93	86	100	75	80	85	1,644	1,274	890	1,269
EX2	71	100	100	90	67	75	100	81	1,752	1,144	755	1,217
EX3	100	93	100	98	100	100	100	100	1,654	1,189	739	1,194
EX4	93	100	100	98	67	100	100	89%	1,662	1,186	665	1,171
EX5	93	86	100	93	100	75	100	92	1,814	1,313	795	1,307
PRO	100	100	100	100	100	100	100	100	384	368	360	371

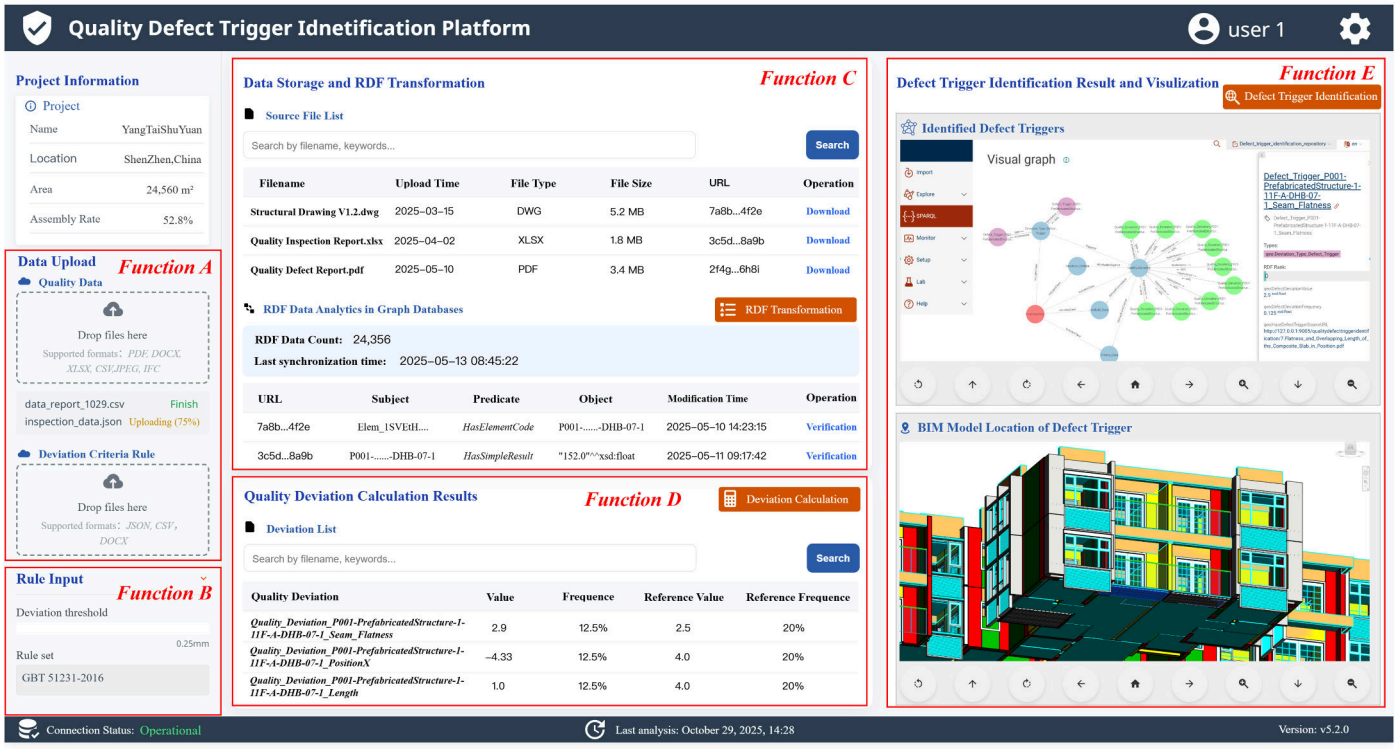


Fig. 18. User-interface design for the quality defect trigger identification system.

and presents the results via a knowledge graph and BIM model views. Additional panels report the system operational status and time-stamp logging. The corresponding interfaces are shown in Fig. 18.

Evaluation

A semistructured interview was conducted with experts in the AECO industry to evaluate the definition of quality defect triggers, the usability and practicality of the proposed framework. The feedback provided valuable insights into the reasoning behind the definition of quality defect triggers, the advantages and disadvantages of the proposed framework, and recommendations for further development. This feedback is expected to contribute to the continuous improvement of the framework's efficiency and effectiveness in future applications.

Six experts from the AECO industry were invited to assess the feasibility of the proposed framework and its implementation prototype by answering five questions. The six experts were purposively selected based on the following considerations:

- They were drawn from key AECO stakeholder organizations and research institutions, including owner organizations, supervision agencies, general contracting companies, and universities;
- The panel combined experienced industry practitioners with highly educated researchers, thereby integrating practical and academic perspectives; and
- All experts had practical or research experience directly related to construction quality management.

Detailed information is provided in Table 3. In this semistructured interview, five questions were set, addressing the definition of quality defect triggers, the advantages and disadvantages of the proposed framework, the feasibility of using the framework in

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Table 3. Detailed information of interviewees

ID	Organization	Position	Education	Working experience
OW1	Owner	Project manager	Bachelor's degree	29 years
SA1	Supervision agency	Supervision engineer	Bachelor's degree	16 years
MC1	Main contractor	Project quality manager	Bachelor's degree	7 years
MC2	Main contractor	Project engineer	Doctoral degree	7 years
UN1	University	Postdoctoral fellow	Doctoral degree	9 years
UN2	University	Postdoctoral fellow	Doctoral degree	8 years

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engineering practice, and suggestions for its future development. The five questions are illustrated in the Table S2.

The interview results indicate unanimous agreement among respondents that quality defect triggers are prevalent in engineering practice, particularly in concrete construction, steel installation, and prefabricated component assembly. Most participants recognized the proposed framework's significant strengths in multisource data integration, defect trigger localization, semantic modeling, and scalability. Nevertheless, they also identified several limitations, including limited visualization of identification results, low integration with mainstream tools such as BIM, suboptimal timeliness, and a strong dependence on high-quality data. Although some interviewees remained cautious due to learning costs and established work habits, the majority expressed willingness to test the prototype in their research work or project management practice.

Suggested improvements include integrating safety management together, deeply embedding BIM models and construction drawing tools, enhancing the detection of management-related and human-related defect triggers, providing corrective recommendations for identified triggers, establishing a matching data-upload mechanism for project management, expanding data sources, optimizing mobile interfaces, and incorporating reinforcement learning or large language models to bolster analytical capability. These recommendations provide a clear roadmap for improving the quality defect trigger identification framework. The responses from interviewees are illustrated in the Table S3.

Discussion

Early studies primarily relied on surveys and document analysis to extract and identify construction quality defects and potential hazards. During this period, the relationships among key concepts, i.e., risk, hazard, deviation, and trigger, were not systematically organized, and many notions preceding defect formation were collectively categorized under the term of causes (Aljassmi and Han 2013; Burati et al. 1992; Josephson and Hammarlund 1999; Ye et al. 2015), reflecting the prevailing understanding in industry practice at the time.

However, within such a broad orientation, the process through which these causes evolve from latent to manifest states was largely overlooked, making it difficult for refined analytical methods and technologies to be effectively applied. This study refined the broad category of causes by distinguishing between deviations and defect triggers, following the causation frameworks established in previous research, while extending the understanding of the underlying mechanisms of quality defect formation. It introduces a primarily physics-based quantitative and information-driven approach that, to some extent, also integrates management-related factors into the analysis.

Current research on quality defect identification has been supported by paradigms from safety management and empowered by emerging technological innovations, exhibiting both diversification and breakthroughs. From the perspective of adopting safety management paradigms, current research has not fully distinguished the differences between quality and safety attributes or the mechanisms that ultimately lead to accidents. As a result, conceptual overlaps and interchangeable uses of terms are common. Further exploration is needed to determine how mature theories and models, such as the Swiss Cheese Model (Larouzee and Le Coze 2020; Reason et al. 2006) and Heinrich's Law (Dunlap et al. 2019; Heinrich 1941), can be adapted and refined to better suit the context of quality management. Therefore, this study discussed the distinction between abrupt (drawing on safety management) and gradual mechanisms in the literature review and evolution mechanism sections. During the development of technical methods, it further explored the coupled effects between gradual quality deviation accumulation and abrupt changes triggered by specific triggers.

From the perspective of technological innovation and adaptive application, existing studies have employed a wide range of technologies and methods, from early BIM and rule-based checking (Kang et al. 2020; Lee et al. 2016), and more recently, deep learning and machine vision (Cha et al. 2024; Guo et al. 2021; Zhong et al. 2025), forming a rich technological foundation. However, these efforts lack an integrated combination of techniques and methods aligned with the evolutionary chain of quality defects. To address this gap, this study adopts an integrated evolution mechanism–ontology–linked data–digital twin framework that semantically connects quality standards, construction processes, and project quality data, with the DT serving as the visualization and interaction platform for early identification of quality defects.

The methodological contribution mainly lies in the development of a reusable workflow for deviation calculation and defect trigger identification. This workflow is implemented via RDF instantiation and SPARQL reasoning, which substantially enhances automation and traceability. Through a three-layer semantic model, mapping templates, and Python-based tools, it standardizes, interoperates, and efficiently integrates multisource heterogeneous quality data.

The application of the proposed framework in the prefabricated composite-slab installation case demonstrated its practical feasibility for early identification of quality defect triggers. Specifically, it entailed detecting a 2.9-mm seam-flatness deviation and an x-axis misalignment trigger event. A comparative experiment with 10 participants showed that the proposed method surpassed the checklist-based approach in both accuracy and efficiency, and expert interviews indicated improvements in data reliability, accountability, and stakeholder collaboration, supporting a shift from reactive to proactive quality management. As a practical implication, the framework may contribute to a shift from reactive acceptance toward more proactive prevention in construction quality management by enabling finer-grained identification of defect triggers and fostering stronger collaboration among stakeholders.

In terms of practical implementation, the framework currently relies on real-time, high-quality data input, highlighting the need for more robust on-site data collection and upload mechanisms. Integration with BIM systems and mobile platforms remains limited, and the visualization layer requires further enrichment to improve user interaction. Additionally, the absence of a standardized defect trigger grading system and corresponding remediation strategies constrains broader adoption. Future work should therefore focus on enhancing data robustness and interoperability, strengthening multiplatform and DT integration, and developing a systematic grading and response model for defect triggers. These improvements will increase the framework's applicability and value for proactive, data-informed quality management in real-world engineering practice.

Conclusion

In the context of the increasing emphasis on quality management and the application of digital technologies in the AECO industry, this study proposed the concept of quality defect triggers and developed a framework for their identification in construction quality management. The research examined the underlying mechanisms of defect trigger evolution, focusing on the coupled effects of gradual deviation accumulation and event-triggered abrupt changes. Building on this theoretical foundation, the study integrated quality data through an evolution mechanism–ontology–linked data–digital twin framework, enabling semantic linkage among heterogeneous data sources.

Furthermore, an iterative deviation-checking approach based on process logic and rule constraints was developed to enhance precision in identifying potential triggers. The framework was implemented within a DT environment to facilitate intuitive visualization, communication, and collaboration among project stakeholders. Finally, validation through a real-world prefabricated composite-slab installation case and a participant-based experiment demonstrated the framework's effectiveness and applicability in supporting proactive, data-informed construction quality issue identification.

Despite the progress achieved in quality defect trigger identification and multisource data integration, several limitations and directions for future research remain. This study clarified the identifying features of defect triggers through analysis of their evolutionary mechanisms, yet deeper exploration is still needed to refine their interrelationships and support more accurate classification and mitigation. The framework's reliance on high-quality, real-time data also revealed the need for more robust on-site data collection, transfer, and integration with BIM and mobile platforms. In addition, visualization and user interaction require further enhancement to improve accessibility and collaboration across stakeholders. Finally, the absence of standardized grading and response strategies for defect triggers constrains practical application. Future work should therefore focus on strengthening data interoperability, expanding DT-based integration, and developing systematic grading and mitigation models to enhance the framework's effectiveness and applicability in proactive, data-driven quality management.

Data Availability Statement

Some or all data, models, or code that support the findings of this study are available from the corresponding author upon reasonable request.

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Author Contributions

Fenghua Tian: Conceptualization; Data curation; Formal analysis; Investigation; Methodology; Validation; Visualization; Writing – original draft. Ningshuang Zeng: Conceptualization; Methodology; Supervision; Validation; Writing – review and editing. Zirui Li: Validation; Writing – review and editing. Qingzi Ge: Supervision; Validation; Visualization. Qian Wang: Conceptualization; Methodology; Resources; Supervision. Qiming Li: Conceptualization; Funding acquisition; Supervision.

Supplemental Materials

Tables S1–S3 are available online in the ASCE Library (www.ascelibrary.org).

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